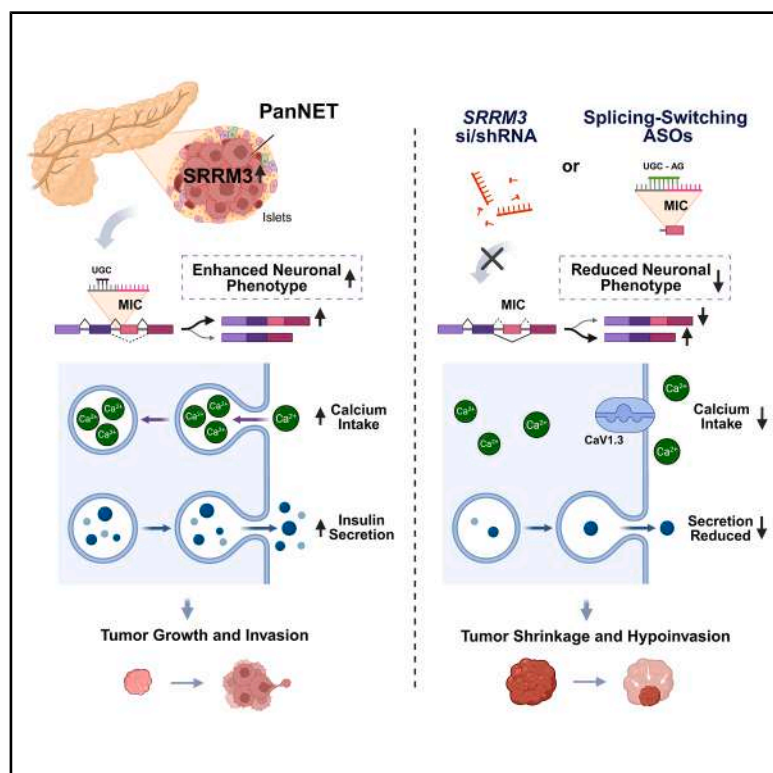


An SRRM3-regulated neural alternative splicing program is subverted to promote tumor progression in pancreatic neuroendocrine cells

Graphical abstract



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In brief

Potiri et al. uncover how the splicing factor SRRM3 drives tumor progression in pancreatic neuroendocrine tumors (PanNETs) via neural microexon inclusion. Aberrant splicing disrupts insulin secretion and calcium signaling, revealing a novel link between neuroendocrine dysfunction and cancer. Therapeutic redirection using splice-switching ASOs can effectively impair tumor growth in preclinical models.

Highlights

- SRRM3 drives a neural splicing program in PanNETs via microexon regulation
- Neural microexons alter insulin secretion, vesicle trafficking, and Ca²⁺ signaling
- SRRM3 knockdown or splice-switching ASOs reduce PanNET growth *in vitro* and *in vivo*
- Targeting microexons with ASOs offers a potent novel therapeutic strategy in PanNETs



Article

An SRRM3-regulated neural alternative splicing program is subverted to promote tumor progression in pancreatic neuroendocrine cells

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SUMMARY

Pancreatic neuroendocrine tumors (PanNETs) are a heterogeneous group of neoplasms arising in pancreatic islets and altering the hormone-secreting function of neuroendocrine cells. Genome-wide approaches have revealed the genomic landscape of PanNETs but have not explained their problematic hormone secretion. We show here that alternative splicing (AS) deregulation is responsible for changes in the secretory ability of PanNET cells. We reveal that the RNA-binding protein SRRM3 is upregulated in PanNETs and favors the inclusion of a group of alternative microexons in certain mRNAs. These microexons are part of a larger neural program regulated by SRRM3, and their inclusion results in protein isoforms that change stimulus-induced insulin trafficking and secretion. By downregulating SRRM3 or inhibiting its binding to three of the microexon-bearing pre-mRNAs in animal and cellular PanNET models, we prove the necessity of SRRM3 for hormone secretion, PanNET progression, and enhancement of the neural component of PanNET tumors.

INTRODUCTION

Pre-mRNA alternative splicing (AS) generates diverse mRNA and protein isoforms from a single gene. Transcripts from over 95% of human multiexon genes undergo AS, and most of the resulting mRNA isoforms are variably expressed across cell and tissue types.^{1,2} Many cell- or tissue-specific and developmentally regulated AS events are controlled coordinately by individual RNA binding proteins (RBPs), and these events are significantly enriched in genes that operate in common biological processes and pathways.^{1,3}

In almost all cancers, the expression of spliceosome components is altered, contributing to the generation of AS isoforms that are advantageous for tumor development.^{4–6} Notably, the impact of the generation of these new isoforms is apparent in almost every hallmark of cancer. Antisense oligonucleotides (ASOs) that prevent the formation of these novel isoforms have emerged as a novel therapeutic approach to splicing modulation that can alter the AS outcome for a specific pre-mRNA.⁷ Preclinical work suggests that ASOs might also have therapeutic value in oncology.^{5,8,9} However, to reach the point of using these or

other tools, one must identify and understand the functions of splice variants that lead to tumor progression.

Pancreatic neuroendocrine tumors (PanNETs) are a rare heterogeneous group of epithelial neoplasms arising from pancreatic islet cells.¹⁰ PanNETs are classified as functional or non-functional, depending on their ability to secrete biologically active hormones and thus elicit characteristic symptomatology. The management of PanNETs is challenging due to the lack of specific biomarkers and efficient treatments; long-term progression-free survival for patients is rare.^{11,12}

Emerging evidence suggests that splicing factors show altered expression levels in PanNETs and thus affect tumor aggressiveness or growth.^{13,14} However, information on disease-related splicing variants affecting PanNET progression is missing. In the present study, we characterized the AS changes in PanNET progression. SRRM3 emerges as a splicing regulator critical for the PanNET characteristic AS changes. Using *in vitro* and *in vivo* models, we showed the significance of AS deregulation for neuroendocrine cell function and tumor growth. Thus, we uncovered the mechanistic details of a neural-specific splicing program that increases the neural component of neuroendocrine



cells, affects their ability to secrete insulin, and drives tumor growth.

RESULTS

Deregulation of AS differentiates PanNET from normal tissues and other cancer types

To investigate changes in mRNA isoform expression between tumors and normal tissue, we profiled the AS patterns of publicly available RNAseq datasets of PanNET patients. The general characteristics of the patients are shown in Figure S1A. Most of the samples were derived from non-functional PanNETs of low grading (G1 and G2). In parallel, we profiled the AS patterns of publicly available RNAseq from normal human islets¹⁵ and normal human pancreatic single cell populations (alpha, beta, acinar, and ductal cells¹⁶). We assessed the expression profiles of neuroendocrine markers for all the samples and verified not only the identity of the normal ones but also the deregulation of the pancreatic cell markers in PanNET samples (Figure S1B). For each splicing event, the “percent spliced in” (PSI) value was estimated, reflecting the fraction of mRNA isoforms derived from a single gene and including the alternative exon, intron, or splice site. A total of 721,551 events, supported by sufficient junction reads to allow their quantification, were identified. Of these events, 239,456 were consistently mapped in at least 70 out of the 129 (62 PanNET and 67 normal) samples analyzed. The majority of these events involved alternative exons (~40%), followed by a large number of intron retention events (34%) and small numbers of Alt3’ (15%) and Alt5’ (~10%) (Figure S1C).

Principal-component analysis (PCA) using the PSI values of the 5000 most variable events revealed distinct clustering of PanNET, islets, alpha/beta, and acinar/ductal cells, indicating the different AS patterns in the respective datasets. This grouping was even more apparent when the PCA analysis was performed with the most variable alternative exon events (Figure 1A), which constitute the majority of mapped events (Figure S1C). Similar PCA analysis of the PSI values of the most variable intron retention events (Figure S1E) did not give distinct grouping for the different tissue/cell categories. Interestingly, the two distinct datasets of PanNET samples that we used^{17,18} clustered very nicely together (Figure 1A). Thus, we focused our analyses on the alternative exon events.

Given that there is still great uncertainty about the cells of origin of PanNETs, we focused mainly on the comparison of alternative exon events between PanNET and normal islets to account for any contribution of all endocrine cell types of the pancreas. Thus, we compared their PSI values in PanNETs with the values in normal islets. Upon filtering for the most significantly changed AS events ($|\Delta\text{PSI}| \geq 15\%$ and p -adjusted value < 0.05), a total of 704 alternative exon events were identified (Table S1). These events appeared to be largely conserved between human and mouse species (Figure S1F). The PSI values of these 704 alternative exons were used to perform an unbiased hierarchical clustering analysis using Euclidean distance (Figure 1B). All the samples were classified into three distinct groups—tumor, islet, and alpha/beta/ductal/acinar—suggesting that the events under study allow differentiation of the distinct

sample types. Furthermore, the larger group of changed events is upregulated (473 alternative exons with positive ΔPSI) in PanNETs compared to normal islets. Correlation analysis was conducted using the Pearson correlation coefficient to evaluate the association between PanNETs and other cancer types from the Cancer Genome Atlas (TCGA) (Figure 1C). PanNETs exhibited a negative linear relationship with all other cancers. This observation supports our hypothesis that AS changes in PanNETs contribute to a distinctive tumoral phenotype (Figure 1C). A similar correlation analysis between PanNETs (the 704 alternative exons) and other NETs revealed the closest similarity of PanNET splicing isoforms with the intestinal NET ones (I-NET; Figure 1D).

Furthermore, we investigated the impact of the changes in the inclusion pattern of the alternative exons on the respective protein isoforms. As shown in Figure 1E, the majority of these events (~54%) are predicted to maintain the ORF and thus result in alternative protein isoforms, while ~30% of the events potentially disrupt the ORF and affect the final protein levels.

Neuronal factor SRRM3 is upregulated in PanNETs, and this is correlated with the upregulation of an islet splicing program

To identify splicing regulators possibly mediating the observed AS changes, we examined the global gene expression data of the PanNET samples for changes in the levels of splicing factor mRNAs in comparison to the normal islet datasets. A stringent analysis led to the identification of 23 splicing factors significantly upregulated in PanNETs compared to normal islets. Of these, seven are known to regulate neuronal splicing (Table S2; Figure 2A).

We then performed a correlation-based splicing factor-event interface analysis, using stringent thresholds to select significantly altered AS events and splicing factors (Figure 2B). Specifically, we calculated Pearson correlation coefficients between the expression levels of each differentially expressed splicing factor and the PSI values of the filtered AS events across all samples. To ensure biological relevance, we retained only strong correlations ($r > 0.6$ or $r < -0.6$). We then quantified, for each splicing factor, the number of splicing events meeting this correlation threshold. Splicing factors with > 100 such correlations were retained for downstream analysis and visualization.

The neural splicing factor SRRM3 emerged among the top-ranked factors with the highest number of statistically significant correlations with AS events. Most importantly, SRRM3 showed the strongest bias among all other splicing factors toward positive correlations (frequency > 150), suggesting that higher SRRM3 expression is associated with increased exon inclusion across a broad subset of events. It also ranked first among all RNA-binding proteins analyzed, based on the total number of correlated splicing events (Figures 2B and 2C). This prioritization was further supported by motif enrichment analysis, which identified the SRRM3-binding motif upstream of a significant number of the 473 events upregulated in PanNETs.

SRRM3 mRNA is enriched in neuronal tissues (particularly in the cerebellum), while its expression levels in the pancreas are very low. In PanNETs SRRM3 mRNA levels increase dramatically, reaching neuronal tissue levels (Figure 2D). A comparative

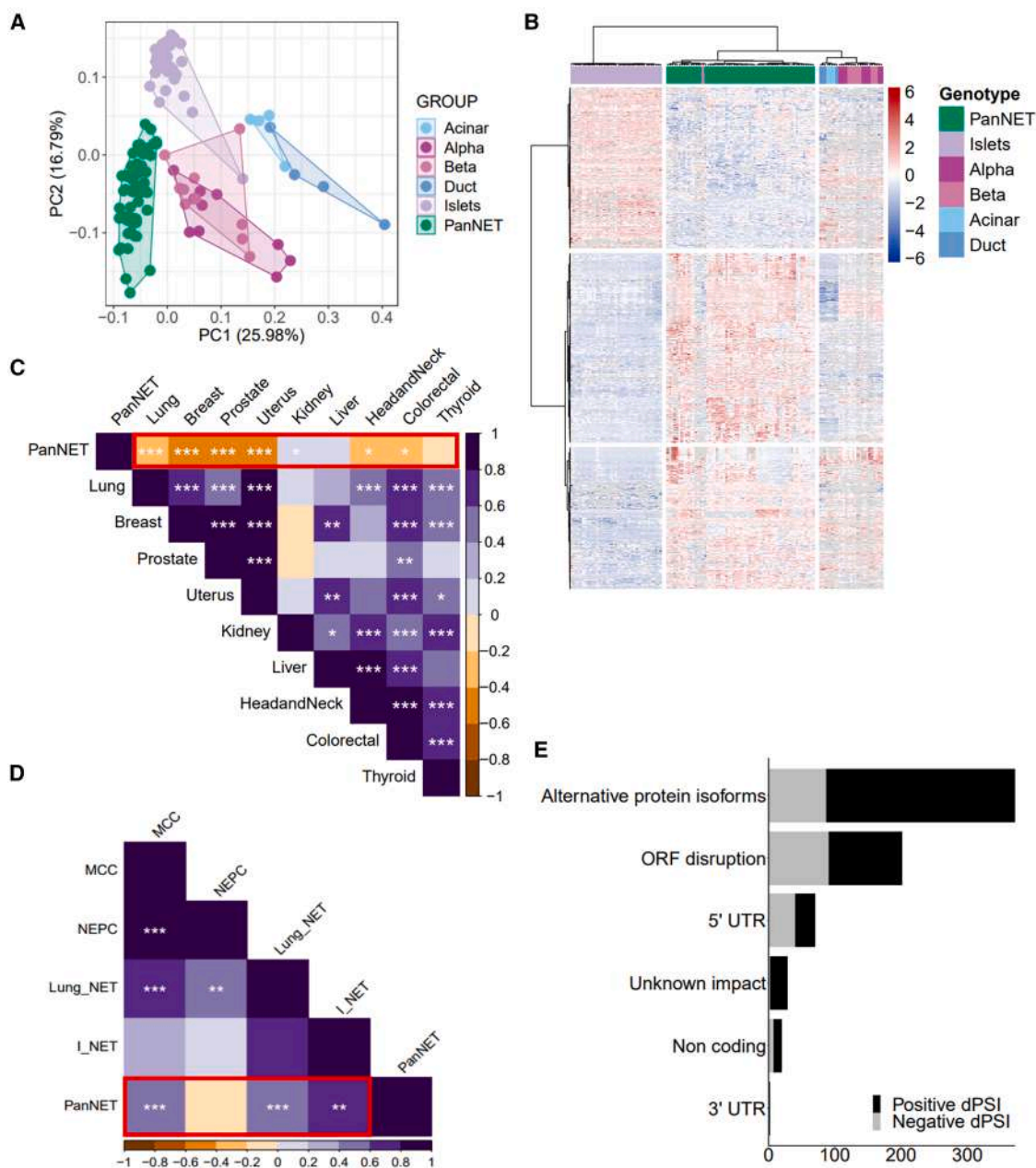


Figure 1. Comparison of alternative splicing in PanNET samples to normal pancreatic tissue reveals the pivotal role of exonic events in neuroendocrine tumors

(A) The principal-component analysis (PCA) illustrates the distribution of the PSI values of the top 5,000 most variable exonic splicing events in tumor samples in comparison with alpha, beta, acinar, and ductal cells or islets. Each point represents an individual sample analyzed.

(B) Heatmap of the top 5,000 most variable exonic splicing events showing the clustering of the samples based on their PSI values, with rows representing the events and columns the samples analyzed.

(C and D) Correlation plots of the common altered exonic events between PanNETs ($|\Delta\text{PSI}| > 15$, p -adjusted value < 0.05) and other cancers from TCGA database (C) or other types of NETs ($|\Delta\text{PSI}| > 5$, p -value < 0.05) (D). Association values determined by Pearson coefficient are visualized using color grading. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

(E) Stacked bar chart showing the predicted impact that the significantly altered exonic events have on the protein level. The x axis shows the number of changing events, whereas the y axis denotes the change caused at the protein level.

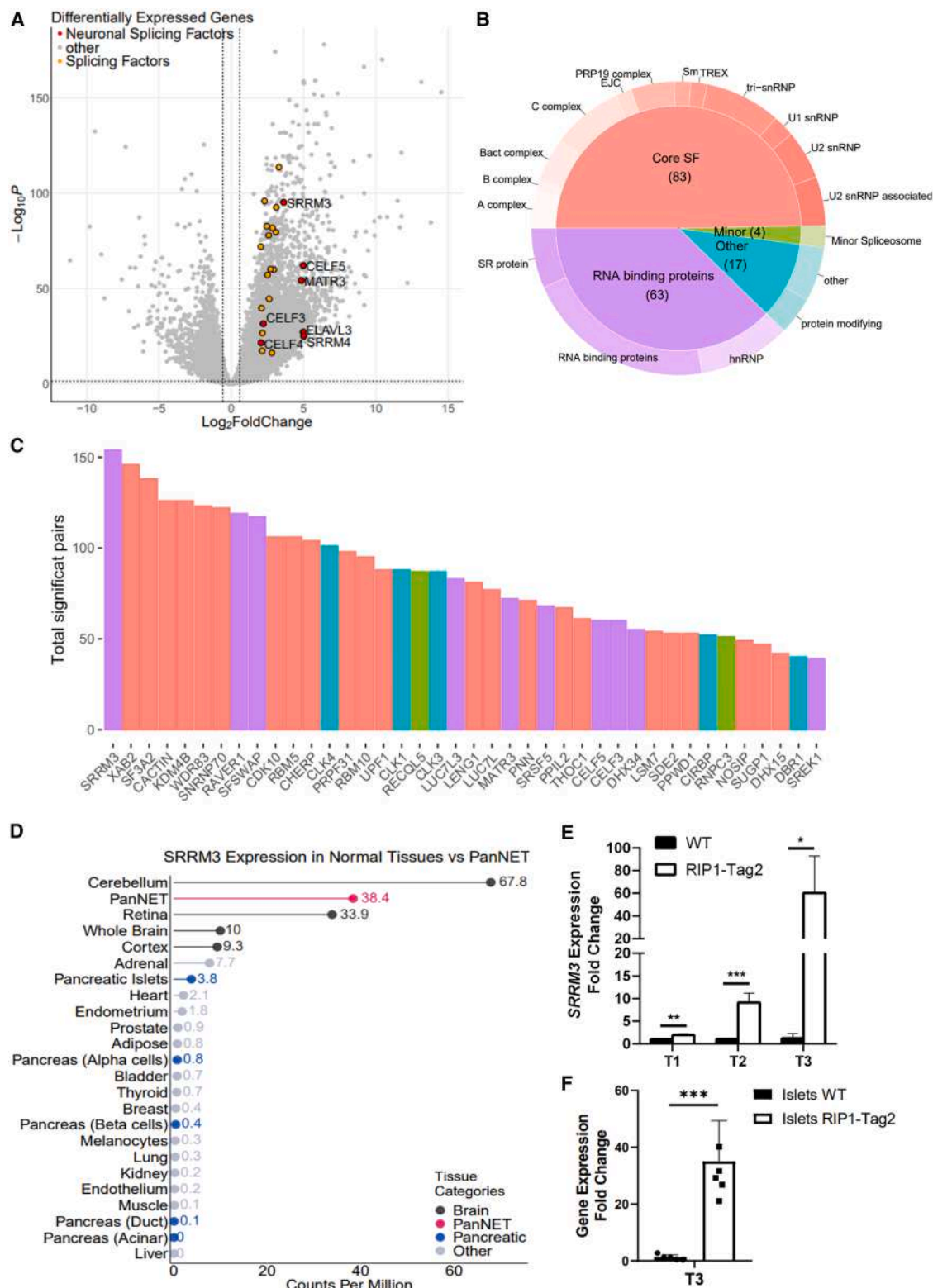


Figure 2. The neuronal splicing factor SRRM3 is significantly upregulated and potentially controls splicing upregulation in PanNETs

(A) Volcano plot of the $\log_2$ fold-change values in the mRNA levels between PanNET samples and normal pancreatic islets. In color, splicing factors significantly upregulated in PanNETs ($|\log_2$ fold-change $| > 2$, p -adjusted value < 0.05). In red, factors regulating neuronal splicing.

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analysis of the deregulation of the *SRRM3* mRNA levels in PanNETs and in different TCGA cancers revealed that in PanNETs, *SRRM3* mRNA exhibits the highest levels of deregulation (Figure S2B). Furthermore, *SRRM3* mRNA levels are higher in the later PanNET stages, III and IV, according to the patients' data analysis (Figure S2C).

We used the RIP1-Tag2 (RT2) mouse model¹⁹ in C57Bl6/J background to assay the levels of *SRRM3* mRNA in PanNETs. PanNETs develop in RT2 mice due to the expression of SV40 T-antigen (Tag) driven by a rat insulin promoter (RIP), leading to rapid and synchronized insulinoma formation in β -cells. Using qPCR, we assayed the expression levels of *SRRM3* mRNA in pancreatic RNA isolated from RT2 mice at three distinct time points: T1 = 8 weeks, T2 = 12 weeks, and T3 = 16–17 weeks (described in STAR Methods; Figure 2E).¹⁹ The blood glucose, insulin, and corticosterone levels of the RT2 mice at T1 are similar to those of their wild-type (WT) littermates (Figure S2D). At T2, islet hyperplasia is accompanied by increased blood insulin levels and reduced glucose levels. At T3, growth of large, insulin-producing tumors is easily detected,^{19,20} together with increased blood corticosterone and insulin levels and further reduction of blood glucose (Figures S2D and S2E), indicating a largely impaired glucose homeostasis state of these mice. Interestingly, *SRRM3* mRNA had a two-fold increase in the pancreas of RT2 mice already at T1, and the overexpression became more and more apparent with tumor progression, reaching a 60-fold increase in the pancreas of mice at T3. *SRRM3* mRNA overexpression was also apparent in pancreatic islets from RT2 mice compared to islets from WT mice (Figure 2F).

SRRM3 participates in neuroendocrine cell dysfunction via regulation of AS of a group of neural microexons

It is known that *SRRM3* controls the splicing of microexons < 27 nts,²¹ particularly in locations where *SRRM4* is not expressed,²² which is the case for a small subset of the PanNETs analyzed (Figure S3A). Recently, *SRRM3* was reported as an important regulator for glucose homeostasis in pancreatic islets, regulating the AS of a conserved subset of microexons²³ in beta cells. Intriguingly, our analysis revealed that the vast majority of the microexons altered in PanNETs (83%) were upregulated, with a broader range of Δ PSI values, compared to other upregulated exon events, thus pinpointing a significant role of microexons in these tumors (Figure 3A). Comparative analysis of the microexon events identified by Juan-Mateu et al.²³ as deregulated upon *SRRM3* knockdown in the pancreatic beta-cell line EndoC-betaH1, with the 179 microexons identified as deregulated in PanNETs by our analysis, revealed a very weak negative correlation between the two microexon groups (Figure S3B).

However, all the microexon events that are common between the two groups appear to be upregulated in PanNETs (Figure S3C). A positive correlation was observed when we compared the 179 microexons that are deregulated in PanNETs with the group of neural (N) and pancreatic/neural (PN) microexons that are deregulated in HeLa cells upon overexpression of *SRRM3* (Figure S3D).²³ Furthermore, the majority of the mRNAs affected by the differential inclusion of microexons in PanNETs are genes that bear microexons important for neurogenesis²⁴ (Figure S3E).

To investigate the biological roles of the deregulated microexons in PanNETs, we analyzed the biological processes (BPs) associated with the affected genes. After using Revigo²⁵ to remove redundant Gene Ontology (GO) terms, axon development, vesicle-mediated transport, synaptic vesicles, and signaling were among the most enriched BPs, pinpointing the deregulation of a neuronal phenotype in PanNETs (Figure 3B). Given that these GO terms signify cellular functions that are critical for neuroendocrine tissue homeostasis, we focused our analyses on the impact of AS in such processes. For this, we specifically pursued the investigation of a representative group of alternatively included microexons that depict high Δ PSI values between PanNET and normal islet samples and are associated with these categories: CADPS2 (calcium-dependent secretion activator 2), PTK2/FAK1 (protein-tyrosine kinase 2/Focal adhesion kinase 1), CACNA1D (voltage-dependent L-type calcium channel subunit alpha-1D), HERC4 (HECT domain and RCC1-like domain-containing protein 4), ERGIC3 (endoplasmic reticulum-Golgi intermediate compartment protein 3), DCTN1 (dynactin subunit 1), and SNX2 (sorting nexin-2) (Table S1; Figure 3C). According to our analyses, the respective alternative microexons are highly included in PanNETs compared to normal islets or other types of pancreatic cancers (Figures 3D and S3F). Importantly, with the exception of SNX2, the selected events are conserved between humans and mice.

To validate the change in the inclusion percentage for the above-mentioned alternative microexons during PanNET progression, we used pancreatic RNA from the RT2 mice at T1, T2, and T3. For all the microexons tested, we could detect increased inclusion already at T2 of the RT2 mice compared to their littermate WT mice and an even more profound increase at T3 (Figure 3E) following the overexpression pattern of *SRRM3* mRNA (Figure 2E). For certain cases, the increased inclusion in RT2 versus WT was obvious even at T1 (Figure 3E, *Cacna1D*, *Ptk2*). Analysis of RNA from individual tumors agreed with the elevated inclusion levels of the microexons in RT2 mice (Figure S3G). The use of RNAs from purified pancreatic islets allowed validation of the increased inclusion levels of the

(B) Pie-donut chart of the splicing factors that were found deregulated in the differential expression analysis ($|\log_2\text{fold-change}| > 0.58$, p -adjusted value < 0.05), organized in classes based on splicing-related proteins.

(C) Bar plot illustrating the number of positive correlations of the deregulated splicing factors from panel B with the significantly altered exonic events from the splicing analysis ($|\Delta\text{PSI}| > 15$, p -adjusted value < 0.05), in a descending order. Colors depict different splicing classes as shown in (B).

(D) Lollipop chart showing *SRRM3* average cRPKM values from the differential expression analysis regarding PanNET and normal pancreatic tissues analyzed in comparison to other normal tissues. Data acquired from VastDB.¹⁴ Color coding corresponds to organized categories as indicated.

(E and F) Bar plot of RT-qPCR analysis of *SRRM3* average expression in WT and RIP1-Tag2 mice at three different time points (T1, T2, T3) (E) or in islets isolated from WT and RIP1-Tag2 mice from the T3 time point. *GAPDH* levels were used for the normalization. Error bars represent $\pm$ SEM. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

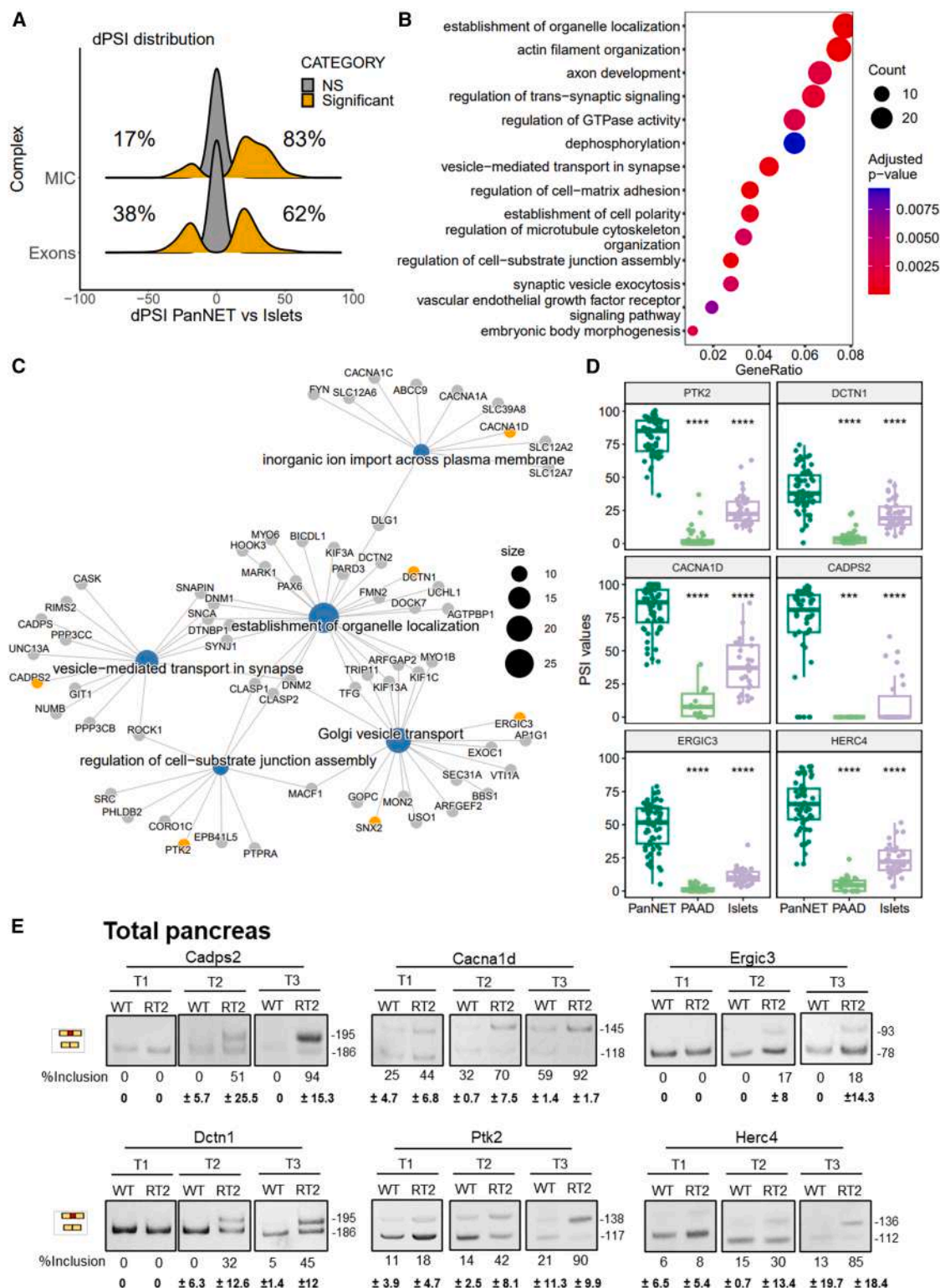


Figure 3. Characterization of a group of highly upregulated microexons in PanNETs compared to normal pancreatic islets

(A) Distribution of Δ PSI values for exonic and MIC events in PanNETs compared to normal islets. In gray: non-significant (NS); in orange: significant events. Percentages indicate the positive or negative Δ PSI values of the events.

(B) Dot plot depicts the results from the Gene Ontology-biological process (GO-BP) enrichment analysis, described in STAR Methods. Color coding indicates the adjusted p value range. Dot size depicts the number of genes involved in each process as indicated.

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microexons in RT2 mice compared to their littermate WT mice, however, with smaller inclusion percentages (Figure S3H). Increased insulin and glucagon levels in the purified islets verified their successful isolation (Figure S3I); however, the reduced inclusion percentages could possibly be explained by the fact that islets derived from tumors are a mixture of normal and hyperplastic islets,²⁶ but could also pinpoint the contribution of non-islet cells in the detected AS changes.

SRRM3 controls the deregulated microexons and the response of intracellular calcium levels to stimuli in PanNET cells

There is functional evidence for the close association of calcium levels with secretory granule exocytosis in normal pancreatic β cells *in vitro*.^{27,28} Interestingly, the microexons that depict the highest Δ PSI between PanNETs and normal islets and are possibly controlled by SRRM3 affect microdomains of proteins that participate in this highly orchestrated process in islets and neuronal cells. For example, the alternative microexon that is almost exclusively included in the *CADPS2* pre-mRNA in PanNETs (exon 19) is a brain-specific neuronal microexon,²⁴ adding three amino acids between two alpha helices in the coiled-coiled domain of the CADPS2 protein (Figure S4A). Importantly, this event has not been detected as altered in other NETs or TCGA cancers (Tables S3 and S4). CADPS2 is of fundamental importance for priming synaptic vesicles for fusion and neurotransmitter release in neurons, large dense core vesicle (LDCV) secretion, and the stability and recruitment of insulin granules in mouse pancreatic beta cells.^{29,30} The significantly high inclusion of the alternative microexon of *CACNA1D* pre-mRNA (exon 44, also a brain-specific microexon²⁴) adds nine amino acids at the C-terminal cytoplasmic region of the L-type calcium channel Cav1.3 (Figure S4A), a protein important for glucose-induced insulin secretion by pancreatic islets.³¹ This results in the expression of a protein with reduced calcium-dependent inactivation capability in mouse brain.³² The inclusion of a microexon in *PTK2* pre-mRNA (exon 6; Figure S4A) results in a FAK1 protein isoform with altered levels of glucose-induced phosphorylation of Tyr397, and this change affects the docking of insulin granules at the plasma membrane of pancreatic beta cells.^{33,34} Additional brain-specific²⁴ microexons are included in PanNETs in the pre-mRNAs of *ERGIC3*, *DCTN1*, and *HERC4*, though their function remains unknown (Figure S4A). The above-mentioned isoforms have not been linked yet to neuroendocrine tumor development.

To verify the dependency of these events on SRRM3, we used the PanNET cell lines, QGP-1, BON-1, and NT-3. QGP-1 cells originate from a primary pancreatic carcinoma, expressing mainly somatostatin and very low levels of insulin.³⁵ BON-1 cell line was established from a functional human pancreatic

carcinoid tumor, and NT-3 from the lymph node of a patient with well-differentiated pancreatic NET, expressing high insulin levels.³⁵ We downregulated the levels of *SRRM3* mRNA in QGP-1 cells by stable transfection with constructs that result in expression of two different shRNAs specific for *SRRM3* or a non-specific (NT) shRNA. The different *SRRM3*-shRNAs used produce varying degrees of *SRRM3* silencing (Figure 4A). Following this observation, for BON-1, we used only the more effective shRNA for *SRRM3* downregulation (Figure S4B). For both cell lines, as for NT-3 cells, the levels of *SRRM3* were also downregulated using specific siRNAs against *SRRM3* and scrambled siRNAs as a control. In all cases, the effectiveness of the downregulation of *SRRM3* mRNA was verified by qPCR (Figures 4A and S4B). At this point, we also assessed whether the *SRRM4* mRNA levels were affected by *SRRM3* downregulation. We detected upregulation of the *SRRM4* mRNA when *SRRM3* is downregulated using shRNAs. When siRNAs were used for the downregulation, we could not detect any significant change in the levels of *SRRM4* mRNA (Figure S4C). These findings suggest that prolonged SRRM3 suppression via shRNA may induce compensatory upregulation of the SRRM4 paralog, a response not seen with the short-term effects of siRNA-mediated knockdown.

Further verification of the downregulation of SRRM3 at the protein level is presented in Figure 4B after Liquid chromatography with tandem mass spectrometry (LC-MS/MS) analysis of extracts of the control and *SRRM3* knockdown derivatives of QGP-1 and BON-1 cell lines. The downregulation of the endogenous SRRM3 is presented compared to GAPDH and Histone H4 as endogenous controls and also in comparison to the abundant paralog of SRRM3, SRRM2, and the unrelated SR protein, SRRM1.²¹ Following verification of SRRM3 downregulation, we performed RT-PCR to assay the inclusion levels of the microexons of interest (Figures 4C–4E). For all three PanNET cell lines, we could detect the expected decrease in inclusion of the microexons upon downregulation of *SRRM3* mRNA.

As presented in Figure 3, SRRM3 is predicted to control the inclusion of microexons that alter the functionality of proteins in secretory vesicle trafficking and exocytosis. This prompted us to assess the involvement of SRRM3 in the secretory ability of PanNET cells in response to stimuli. For this and given that Cav1.3 is an L-type voltage-gated calcium channel, highly expressed in pancreatic cells, and CADPS2 binds to Ca^{2+} to function in secretory vesicle exocytosis, we first tested whether SRRM3 is critical for changes in intracellular calcium levels upon stimulation by glucose or KCl in PanNET cell lines *in vitro*.

More specifically, NT-3 cells transfected with *SRRM3*-specific siRNAs or with control siRNA (Figure 4A) were incubated with 20 mM glucose, and the intracellular calcium levels were

(C) Gene-concept network (cnetplot) of genes across highly enriched GO-BPs in PanNETs compared to islets. Color coding reflects the genes of interest. The size of the BP nodes reflects the number of genes with altered AS that belong to this BP.

(D) Box plots of the PSI values for each selected splicing event. Each point represents an individual sample analyzed. *p* values were calculated using a one-way ANOVA test. *** *p* < 0.001 and **** *p* < 0.0001. PAAD: Pancreatic adenocarcinoma.

(E) Analysis by RT-PCR and gel electrophoresis of the indicated AS events using RNAs isolated from total pancreatic tissues of WT and RT2 mice, at the time points T1, T2, and T3. The %inclusion values represent the quantification of the inclusion band relative to the total signal from both the inclusion and skipping bands. Error bars, based on the standard deviation ($\pm$ SD) values from the corresponding biological replicates, are depicted below. Product lengths (bp) are marked on the right of each picture.

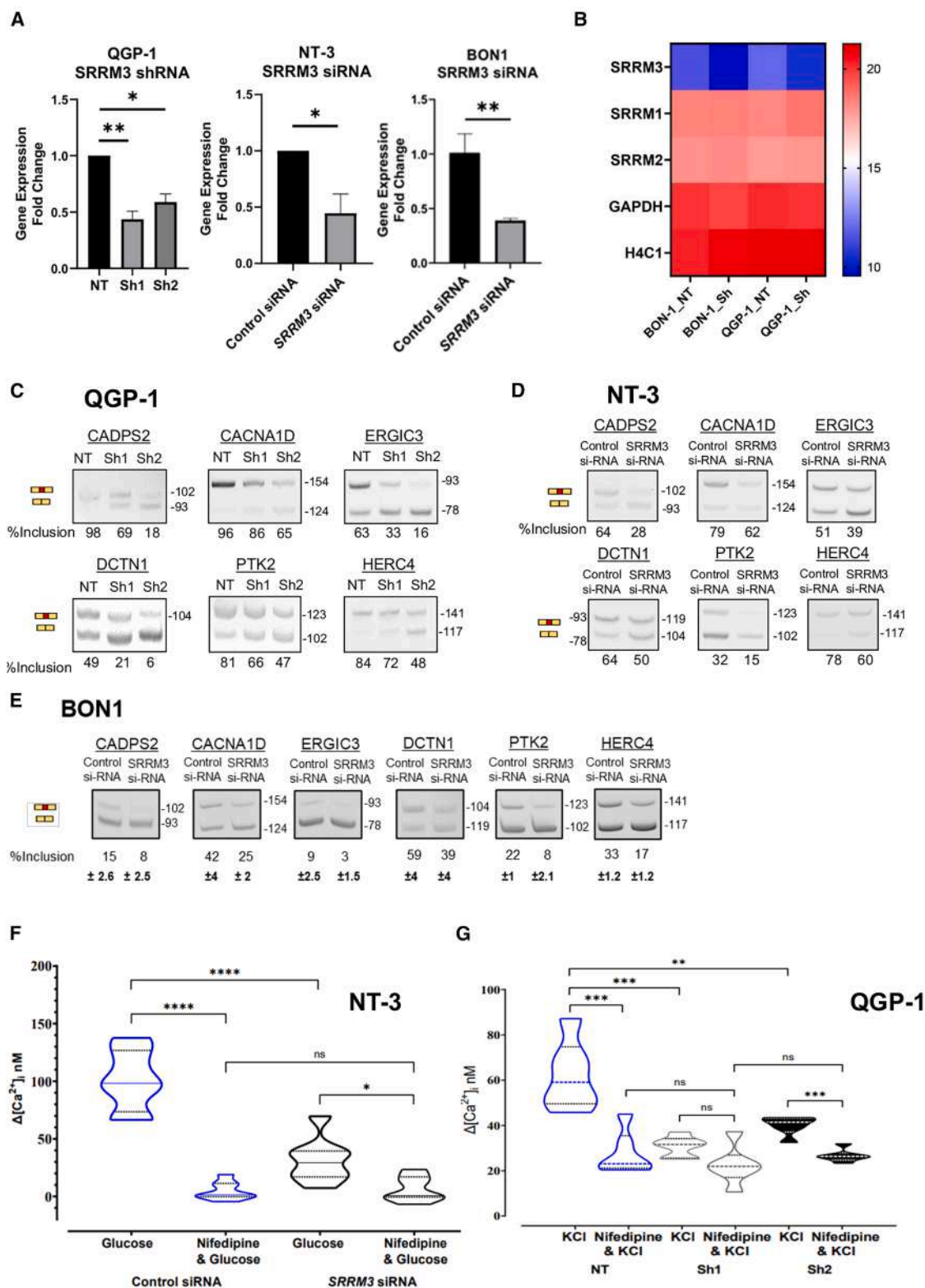


Figure 4. SRRM3 controls microexon inclusion in PanNET cell lines, regulating intracellular calcium responses

(A) Bar plot of RT-qPCR analysis of *SRRM3* average expression in QGP-1, BON-1 and NT-3 cell lines using either *SRRM3*-specific shRNAs (Sh1, Sh2) or *SRRM3*-specific siRNA. *GAPDH* was used for normalization. Error bars represent $\pm$ SEM. * $p < 0.05$ and ** $p < 0.01$.

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assayed using Fluo-4/AM. Control cells responded to the presence of glucose by exhibiting a robust increase in $[Ca^{2+}]_i$ (Figure S4D). In the presence of 10 μ M nifedipine, an inhibitor of L-type voltage-gated calcium channels, the increase in $[Ca^{2+}]_i$ was eliminated (Figure S4E). NT-3 cells with reduced *SRRM3* levels showed a substantially attenuated response to the glucose stimulus (Figure S4D). The small response of NT3-KD cells to glucose, probably attributed to the incomplete downregulation of *SRRM3* in these cells, was eliminated by the addition of 10 μ M nifedipine, which once again completely abolished the response of these cells to 20 mM glucose (Figure S4E). Plotting the difference between the $[Ca^{2+}]_i$ before and after the addition of the stimulant in the NT3 control and *SRRM3* downregulated cells (Figure 4F) clearly showed that a decrease in *SRRM3* mRNA expression leads to a much smaller ability of these PanNET cells to increase intracellular calcium in response to glucose (Figure 4F).

QGP-1 cells with reduced expression of *SRRM3* and control cells were assayed for $[Ca^{2+}]_i$ using KCl to depolarize the cells, because QGP-1 does not respond greatly to glucose stimulation.³⁶ The addition of 50 mM KCl in the cells resulted in a sharp and robust spike in $[Ca^{2+}]_i$ for all three cell lines before $[Ca^{2+}]_i$ leveling off (Figure S4F). The highest increase was recorded in the control QGP-1 cells. A close inspection of the observed spikes in $[Ca^{2+}]_i$ (Figure S4F) revealed that in QGP-1 cells, the increase in $[Ca^{2+}]_i$ reached a peak at 11 s after addition of KCl, whereas in QGP-1-sh1 cells, the spike in $[Ca^{2+}]_i$ was significantly smaller and peaked at 7 s. Also, in QGP-1-sh2, the spike was smaller and peaked at 9 s after KCl addition (Figure S4F). Once again, the robust spikes in $[Ca^{2+}]_i$ generated by KCl were eliminated in all three cell populations in the presence of 10 μ M nifedipine. (Figure S4F). The effect of *SRRM3* downregulation was visualized better after plotting the difference between the $[Ca^{2+}]_i$ before and after the addition of the stimulant in the QGP1-siRNA control and *SRRM3* downregulated cells (Figure 4G). It was clear that the decrease in *SRRM3* expression leads to a much smaller ability of these PanNET cells to increase intracellular calcium in response to glucose. Interestingly, blockage of Cav1.3 in PanNET cells results in reduced viability, as was detected by MTT assays in QGP-1 cells treated with nifedipine (Figure S4H). Together, these results suggest that in PanNET cells, *SRRM3* promotes the formation of a splicing variant of the voltage-gated Ca^{2+} channel Cav1.3, which exhibits altered response kinetics upon cellular stimulation and supports cell viability.

***SRRM3* controls insulin granule exocytosis in PanNET cells**

The reduced increase of intracellular Ca^{2+} levels upon *SRRM3* downregulation in PanNET cells and the essential role of Ca^{2+} influx through voltage-gated Ca^{2+} channels in insulin exocy-

toxis³⁷ prompted us to investigate the impact of *SRRM3* on secretory granule exocytosis. We used the ratiometric reporter, RINS1,³⁸ a double-fluorescent-fusion (superfolder GFP [sfGFP] and mCherry) construct of proinsulin that allows imaging and quantification of insulin secretion. Stimulation of cells transfected with RINS1 by glucose results in an increase of intracellular calcium levels, fusion of mature secretory granules with the plasma membrane, and release of sfGFP-tagged insulin (Ins-sfGFP), whereas the inactive C-peptide, tagged with mCherry, remains intracellular. We transfected *SRRM3*-downregulated NT-3 cells (*SRRM3*-siRNA) or the control cells (control siRNA) with RINS1 and used confocal microscopy to observe the secretory granules (Figure 5A). In unstimulated cells, we could detect the intracellular RINS1 reporter (colocalization of the two fluorescent signals) as expected. Upon stimulation of control cells, a clear increase in the intracellular mCherry signal and separation from the sfGFP one were observed after 10 min, indicating the presence of insulin-sfGFP in granules preparing to be released/and or released from the cells, while mCherry-C peptide remained in the cells. However, in *SRRM3*-siRNA-treated cells, there was no clear separation between the two signals (mCherry and sfGFP), and the cells resembled the unstimulated ones, indicating impaired granule maturation and/or exocytosis. Quantifying the fluorescent signals in the cell supernatants verified the overall reduced ability of the *SRRM3* KD cells to secrete insulin upon stimulation (Figure 5B). The difference in the secretion levels of insulin-sfGFP was confirmed by western blot analysis (Figure S5A) and by insulin ELISA (Figure 5C).

Knowing that the dynamic regulation of Golgi morphology is critical for the transport and packaging of secreted molecules into nascent secretory vesicles and for the progression and metastasis of cancers,³⁹ and given that many of the *SRRM3* regulated microexons alter the function of components of the Golgi vesicle transport process (Figure 3C), we examined the morphology of Golgi upon *SRRM3* downregulation and in response to stimulation by glucose. *SRRM3* downregulation resulted in the reduction of the cellular area covered by the Golgi (Figures 5D, 5E, and S5B), changing the Golgi fragmentation level. Furthermore, stimulation of the cells by glucose induced significant changes to the Golgi morphology only in the presence of *SRRM3*. This finding supports the participation of *SRRM3* in the process of glucose-induced vesicle-mediated transport.

***SRRM3* supports PanNET growth *in vitro* and *in vivo* via regulation of a small number of microexons**

To assay the importance of *SRRM3* on PanNET growth, we first assayed the proliferation of BON-1 and QGP-1-derived cell lines. All *SRRM3* knockdown cell lines showed reduced proliferation rates compared to the control cells (Figures 6A and S6A).

(B) Heatmap depicting the comparative levels of the indicated proteins, as they were identified by LC-MS/MS analysis. Triplicates were used for each condition and each cell type. The median protein intensity values were used.

(C–E) RT-PCR analysis of the indicated events in RNA samples derived from QGP-1 control non-treated cells (NT) and *SRRM3* knockdown QGP1 (Sh1, Sh2) (C), from NT-3 control or NT-3 *SRRM3* knockdown cells (D), and from BON-1 control cells or BON-1 *SRRM3* knockdown cells (E).

(F and G) Violin plots summarizing the differences in $[Ca^{2+}]_i$ ($\Delta [Ca^{2+}]_i$) at the start and end of intracellular calcium measurements shown in Figures S4B–S4G ns > 0.05; * p < 0.05, ** p < 0.01, *** p < 0.001, and **** p < 0.0001.

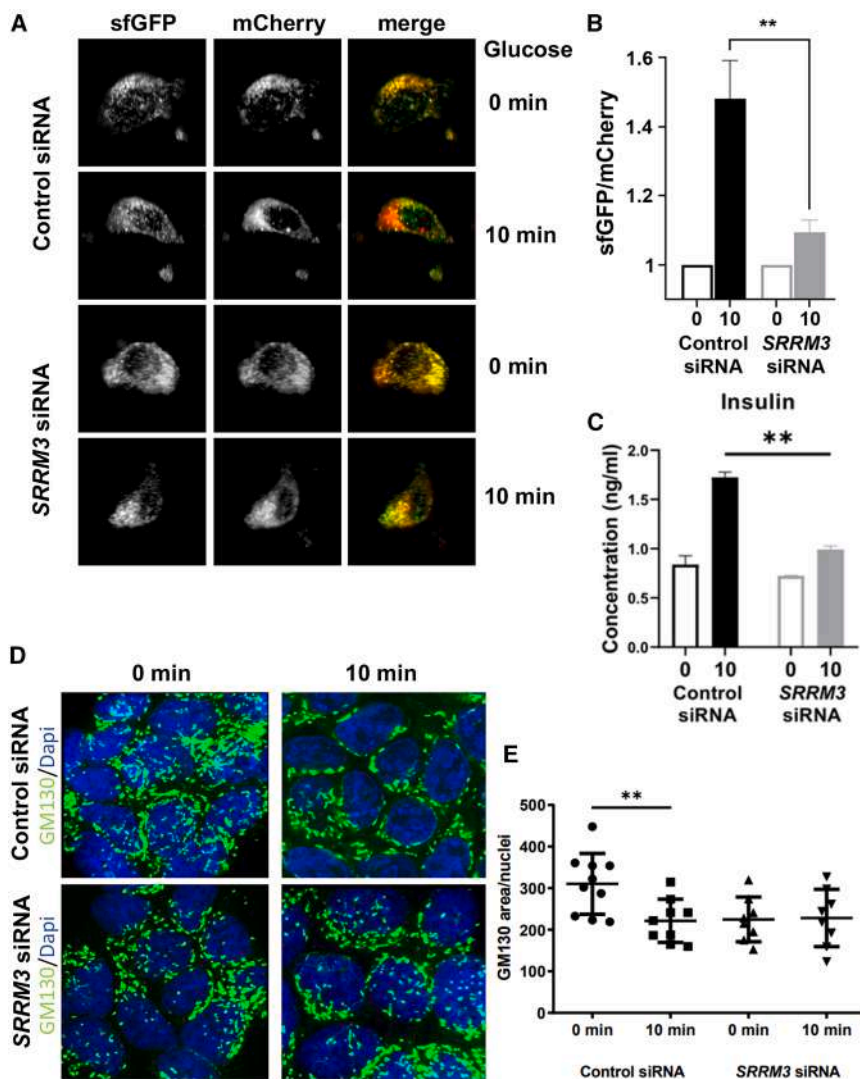


Figure 5. SRRM3 downregulation impairs insulin secretion and alters the Golgi morphology in PanNET cells

(A and B) NT-3 cells co-transfected with control or *SRRM3*-siRNA and pRINS1 were stimulated with 20 mM glucose/30 mM KCl, as indicated. (A) Representative confocal single plane images of 30 cells presented in (B). (B) Graph showing the mean fold-change in the sfGFP/mCherry signal 10 min after stimulation relative to T0 of three independent experiments. Error bars represent $\pm$ SEM. ** $p < 0.01$.

(C) Quantification of insulin levels in the NT-3 cells used in (A) and (B) using a human insulin ELISA kit. (D and E) QGP-1 cells transfected with either control or *SRRM3*-siRNA, stimulated with 20 mM glucose/30 mM KCl, as indicated, were stained with antibodies against GM130 Golgi marker and DAPI. (D) Representative confocal images of cells. (E) Graph presenting the mean Golgi/nuclei area (ten cells from each category were quantified). ** $p < 0.01$.

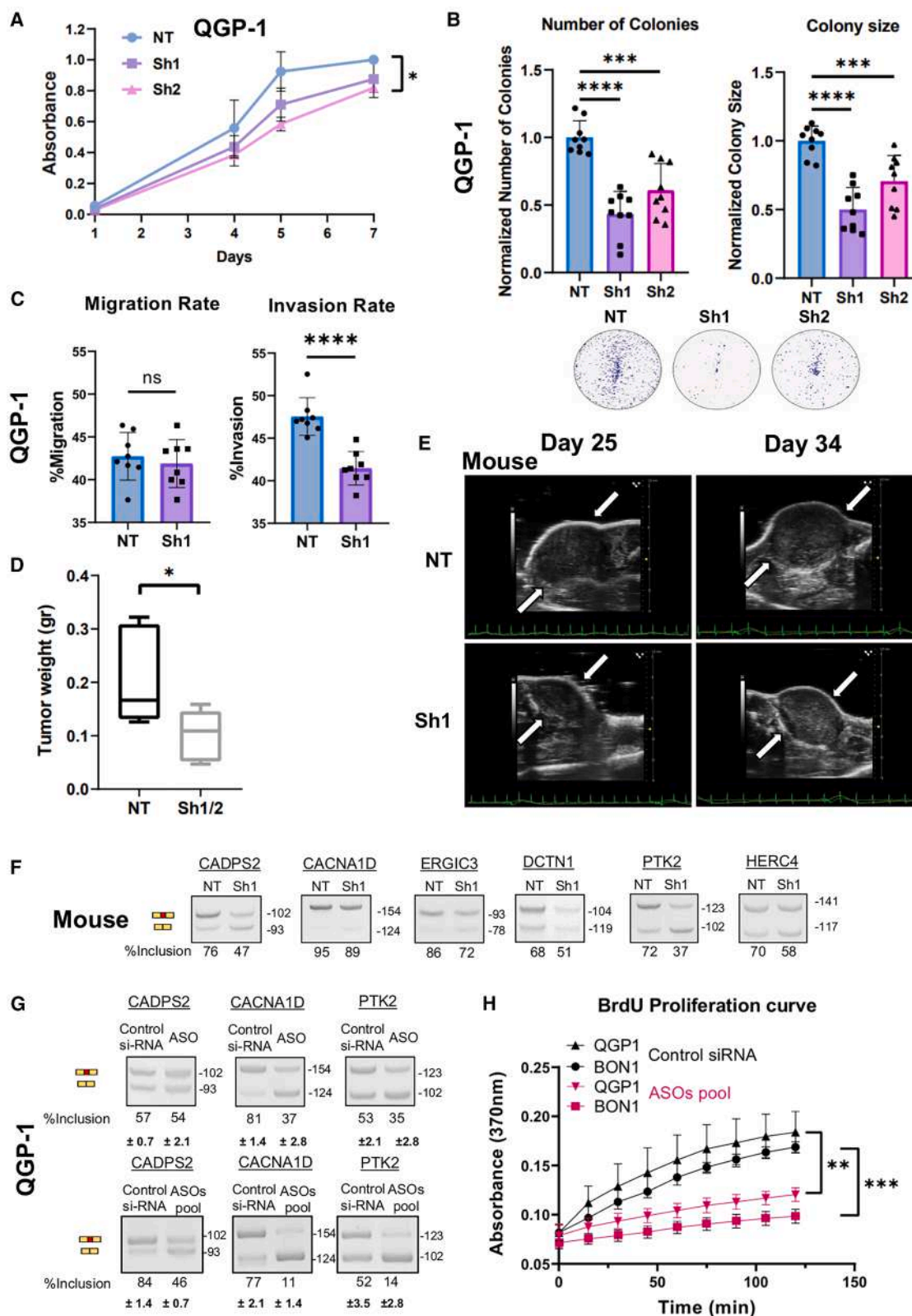
we monitored tumor growth by ultrasonography during the course of the experiment (representative images in Figure 6E). At the endpoint, RNAs from tumors were isolated and used to verify the reduced levels of *SRRM3* mRNA and the expected effect on the inclusion of microexons regulated by *SRRM3* (example in Figure 6F).

To verify that the effect of *SRRM3* is via its ability to regulate the inclusion of neural microexons, we tested whether deregulation of individual microexons was sufficient to alter the PanNET characteristics that depend on *SRRM3*. For this, we designed antisense oligonucleotides (ASOs) harboring 2'-O-methyl

phosphorothioate modifications targeting microexon 3' splice sites and the adjacent regulatory sequences, including the UGC motif required for *SRRM3*-dependent control of microexon inclusion.²³ We initially selected the microexons of *CADPS2*, *PTK2*, and *CACNA1D* as the most deregulated microexon events among the validated ones and because they affect critical components of calcium-induced secretory pathways.^{31,41,42} In all three cases, introduction of the ASOs in QGP-1 or BON-1 cells, in comparison to control siRNAs, induced microexon skipping, as expected. Importantly, the effect of ASOs against the *CACNA1D* and *PTK2* microexons was very similar to the effect of *SRRM3* downregulation on these microexons (Figure 6G). However, using a pool of the three ASOs resulted in a better skipping for all three microexons (Figure 6G, lower panel and S6G). Following this, we tested the effect of these ASOs on PanNET characteristics. It was prominent that upon treatment with the pool of ASOs, the formation of the characteristic PanNET cell spheres was reduced (Figure S6H). The proliferation potential of the cells treated

Moreover, we detected the reduced ability for colony formation upon *SRRM3* downregulation (Figures 6B and S6B). A change in the ability of QGP-1 cells to form spheres was also noted (Figure S6C), supporting the dependency of these cells on *SRRM3* for the development of PanNET phenotypic characteristics. Furthermore, we tested the metastatic potential of these cell lines using transwell migration and invasion assays.⁴⁰ Interestingly, downregulation of *SRRM3* did not alter the ability of both QGP-1 and BON-1 cells to migrate toward the FBS-containing medium. However, the ability of the cells to invade into a Matrigel substrate relies largely on the presence of *SRRM3* for both cell lines (Figures 6C and S6D).

Subsequently, *SRRM3*-knockdown and the respective control QGP-1-derived cell lines were used in xenotransplantation experiments in NOD-SCID mice. Tumors derived from cells with downregulated *SRRM3* showed significantly reduced growth (Figure S6E). At the endpoint, the significantly reduced tumor weight due to *SRRM3* downregulation was measured (Figure 6D). For accuracy and to keep the tumor volumes small,



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with the ASOs was also assessed using a BrdU incorporation assay. Cells transfected with the pool of ASOs for 72 h showed a reduced proliferation rate compared to the control QGP-1 or BON-1 cells (Figure 6H). Using MTT assay, 72 h after transfection, we tested whether the increased skipping induced by the ASOs affected the viability of QGP-1 cells. As shown in Figure S6I, a significant reduction in the viability of the PanNET cells was detected as a result of the increased skipping of the microexons of *CADPS2*, *CACNA1D*, and *PTK2*.

SRRM3-regulated microexons promote the increase of the neural component of neuroendocrine cells in PanNETs

Given that the majority of microexons deregulated in PanNETs are classified as “neural” and are important for neurogenesis (Figure 3C) and that high intratumoral nerve density correlates with tumor progression,⁴³ we assessed whether the human PanNET samples we analyzed exhibit increased neural components. Using unbiased hierarchical clustering (Euclidean distance), we generated a heatmap of the expression levels of genes characterized as neural⁴⁴ (Figure 7A). All samples were grouped in one islet and two tumor clusters, suggesting that the genes under study allow differentiation of the distinct sample types. In addition, tumors of larger size appear more distant from the normal islets than the smaller tumors. Furthermore, the majority of the neural genes were found to be generally upregulated in PanNET samples, compared to the normal islets, suggesting an increase of the neural character of PanNETs (Figure 7A). To further examine this, we assessed the levels of selected neural markers in the RT2 mouse model at the distinct time points during PanNET progression (Figure 7B). *Tubb3* and *Pgp9.5* are neuronal markers expressed mainly in the brain and to a lower level in normal pancreas.⁴⁵ The levels of both *Tubb3* and *Pgp9.5* were upregulated during cancer progression, indicating an increased neural character in PanNETs of the RT2 model, agreeing with the results from the analysis of human PanNET samples (Figure 7A). Following this, we extended our analyses to additional neural genes, namely *NRGN1*, *NGN3*, and *NCAM1* (Figure S7A). These genes have very low levels of expression, if any, in pancreatic cells and are specific for brain cells;⁴⁵ however, they are upregulated in PanNET samples (Table S6). Once again, increased levels of these neural genes

were detected in the pancreas of the RT2 mice at the latest PanNET stage tested.

To determine the role of SRRM3 in the microexon splicing program on the development of the neural genotype in PanNETs, we initially used our xenograft mouse model (Figure 6). For *TUBB3*, we detected a significant reduction in both mRNA and protein levels in tumors developed by cells with downregulated *SRRM3* (Figures 7C and 7D). The mRNA levels of all the neural genes tested were significantly reduced in tumors generated from the QGP-1 cells with downregulated *SRRM3* (Figures 7D and S7B). This result indicates the significance of SRRM3 for the development of the neural phenotype detected in PanNETs.

To further link the splicing activity of SRRM3 on the alternative microexons to the neural character of PanNET cells, we assessed the levels of the same neural markers in QGP-1 and BON-1 cells treated with either *SRRM3*-siRNAs or the pool of ASOs (Figure 6), in comparison to control cells. Intriguingly, the use of the pool of ASOs that prevent the inclusion of the microexons of the *CADPS2*, *PTK2*, and *CACNA1D* resulted in significant downregulation of all the neural markers tested, even more significantly than the downregulation of *SRRM3* alone (Figures 7E and S7C). Overall, these results support the importance of SRRM3 and the alternative microexon splicing program in the development of a neural genotype in PanNETs.

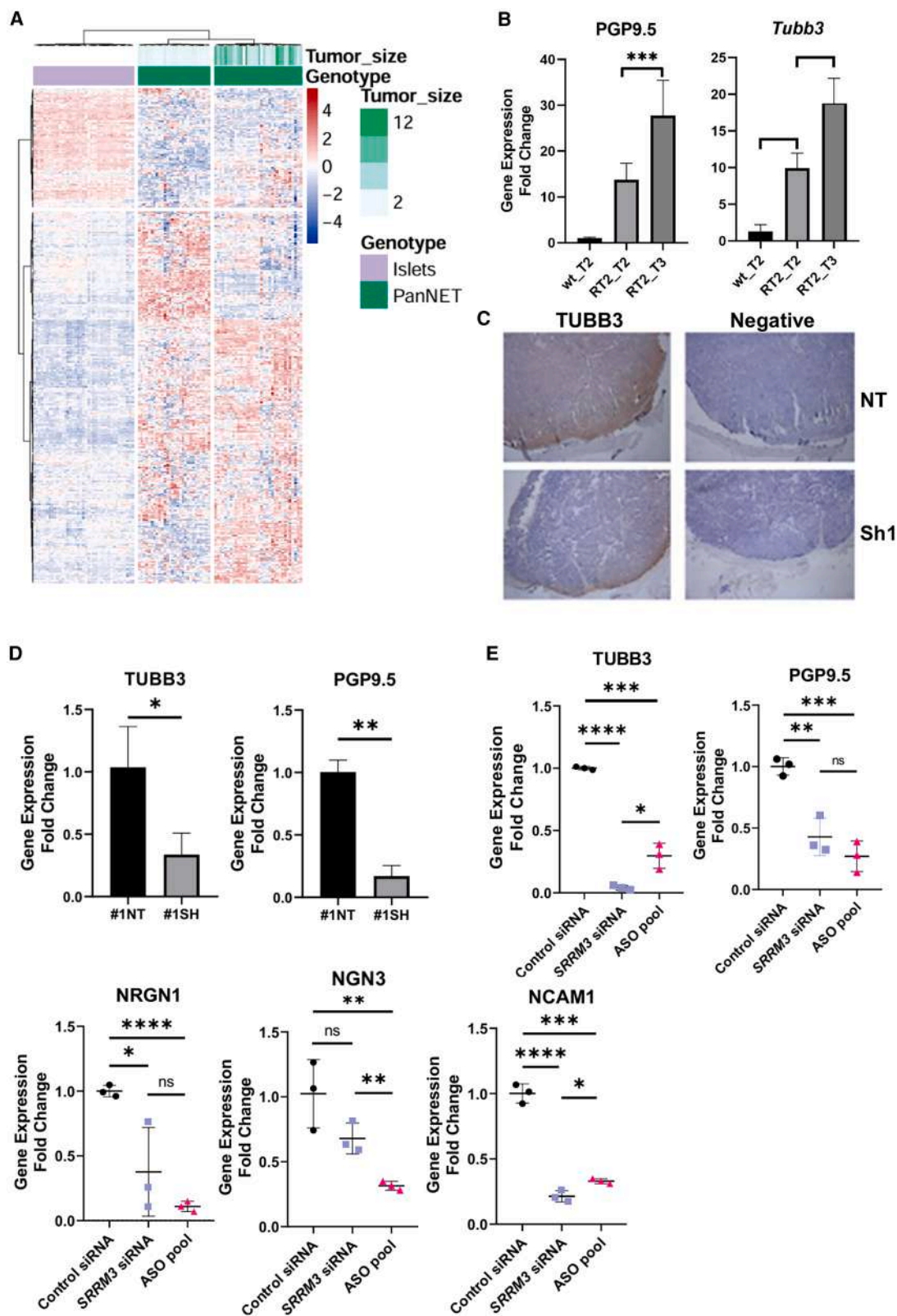
DISCUSSION

In the present study, we show that the deregulation of AS in PanNETs results in the development of a neural phenotype in these tumors. We pinpoint the regulatory splicing factor *SRRM3* as an unexplored source of this phenotype, controlling the inclusion levels of a number of microexons in PanNETs, the majority of which have been characterized as neural microexons.²⁴ These microexons code for surface-accessible, charged amino acid stretches that overlap or lie near functional protein domains, thus contributing to the remodeling of protein-interaction networks that operate during neuronal maturation.²⁴

Interestingly, in PanNETs, deregulation of microexons, driven by the overexpression of SRRM3, results in the alteration of the secretory abilities of neuroendocrine cells. Most of the altered microexons in PanNETs display high inclusion levels in pre-mRNAs associated with axon development, regulation of

Figure 6. SRRM3 promotes tumor growth *in vitro* and *in vivo*

(A) Growth curve for QGP-1 cells treated either with non-specific (NT) or *SRRM3*-specific (Sh1, Sh2) shRNAs. * $p < 0.05$.
(B) Colony formation assay of QGP-1 NT, Sh1, and Sh2 cells. Bar plots of the quantification of colony numbers and sizes are depicted as shown. *** $p < 0.001$ and **** $p < 0.0001$.
(C) Bar plots representing %migration and %invasion rates of QGP-1 NT or Sh1-treated cells, expressed as percentages. Data represent mean $\pm$ SEM from 32 independent fields of areas and triplicates of total inserts in each condition. ns, $p > 0.05$; **** $p < 0.0001$.
(D and E) QGP-1 NT, Sh1, and Sh2 cells were subcutaneously injected into the left (NT) and the right flank (Sh1/2) of NOD-SCID mice, and tumors were monitored weekly over a period of 34 days using the Vevo 3100 Imaging System (FUJIFILM VisualSonics). (D) graph of the tumor weights at the endpoint. * $p < 0.05$.
(E) Representative 3D ultrasound images of mouse No.1 from time points T2 and T3. Arrows delineate the upper right and lower left corners of the tumors. Electrocardiograms (ECGs) are presented at the bottom of the images in green. Size scale bar is in mm.
(F) Analysis by RT-PCR and gel electrophoresis of the indicated splicing events in mouse No.1. Molecular lengths (bp) are marked on the right of each picture.
(G) Analysis as in F of the indicated splicing events in RNA samples derived from QGP-1 control (scramble siRNAs) and QGP-1 cells treated with specific ASO for each splicing event or ASOs pool as a combination of the three specific ASOs.
(H) Graph of continuous measurements of absorbance at 370 nm using ELISA BrdU Kit for QGP-1 and BON-1 cells 48 h after transfection with pool of ASOs in comparison to control siRNAs. Error bars represent $\pm$ SEM. ** $p < 0.01$ and *** $p < 0.001$.



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trans-synaptic signaling, and vesicle-mediated transport and exocytosis. Involved in such functions, the alternative micro-exon incorporated in *CADPS2* mRNA alters a domain responsible for the interaction of this protein with the dopamine receptor DRD2. Alternatively spliced variants of the *CADPS2* mRNA have been connected to the autism spectrum disorder⁴¹ and are subjected to further analyses on their involvement in neurodegenerative diseases.⁴⁶ The increased inclusion of the micro-exon in *CACNA1D* mRNA affects the ability of the L-type voltage-gated Ca^{2+} channel Cav1.3 to regulate intracellular calcium levels (this study), affecting the trafficking of secretory vesicles⁴⁷ and the viability of PanNET cells (this study). Similarly, the microexon of *PTK2* results in the main protein isoform in forebrain neurons,³³ depicting increased autophosphorylation levels and decreased phosphorylation by Src, thereby potentially affecting insulin granule exocytosis^{34,42} and Golgi structure and integrity.⁴⁸ Similarly, increased inclusion of the alternative microexon of *ERGIC3* inserts five amino acids at the COPII-ERV domain, potentially affecting the functionality of ERGIC3 in processing in the ER and/or sorting of molecules into COPII vesicles for transport to the Golgi.⁴⁹ *HERC4* (HECT Domain and RCC1-Like Domain 4) encodes a protein involved in protein trafficking, distribution of cellular structures, and cellular organization.⁵⁰ *HERC4* interacts with USP8 protein, co-regulating the morphology of the endosome and is involved in membrane trafficking at the early endosome stage.⁵¹ *DCTN1* codes for the largest subunit of the dynactin complex, an activator of the molecular motor protein complex dynein.⁵² *DCTN1* plays a key role in dynein-mediated retrograde transport of vesicles and organelles along the microtubules⁵³ and also promotes motor-neuron synapse stability.⁵²

It is not straightforward to assume that overexpression of *SRRM3* alters the ability of neuroendocrine cells to produce hormones, as its role, at least in PanNETs, is probably more complicated. The fact that *SRRM3* downregulation significantly affects the ability of PanNET cells to proliferate, *in vivo* and *in vitro*, suggests that its role extends beyond hormone secretion. It is known that cancer cells release diverse neurotrophic factors and exosomes to promote tumor innervation.^{54,55} Cancer prognosis is related to intratumoral nerve infiltration, especially in organs like pancreas, which display high innervation.^{43,54} On the other hand, vesicle-mediated trafficking and endocytosis and exocytosis are important mechanisms facilitating intratumoral-cell communication.^{39,56} Our results suggest that *SRRM3* and the regulated-microexons have a significant role in the majority of the above-described cellular processes and could be regarded as targets for the development or repurposing of drugs targeting tumor innervation and secretion.

Limitations of the study

In the present study, we show that the mRNA levels of a neural splicing factor, *SRRM3*, are increased in PanNETs and regulate the inclusion of a small group of microexons that control pancreatic islet cell functions. We have verified the downregulation of the *SRRM3* mRNA and protein in our cellular assays; however, the upregulation of the *SRRM3* protein in PanNETs, even though expected by the respective mRNA upregulation, has not been verified due to the lack of specific antibodies. In the present study, we have neither specifically addressed the role of *SRRM3* in K^{+} -ATP-dependent and -independent insulin secretion pathways nor assessed the contribution of the *SRRM3* paralog, *SRRM4*. This work is ongoing. Importantly, for our analyses, we have used human PanNET cell lines and one animal model of PanNETs. The cell lines do not recapitulate accurately the *in vivo* conditions in humans. The animal model is closer, but at the same time, the tumor is driven by an oncogene, which is not the case in human PanNETs. Even though our initial findings have been derived from the bioinformatic analysis of data from a large number of PanNET patients, further analyses of the splicing patterns and changes using human organoids and human tissues would be beneficial.

RESOURCE AVAILABILITY

Lead contact

Requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Panagiota Kafasla (kafasla@fleming.gr).

Materials availability

This study did not generate any new unique reagents.

Data and code availability

- The study does not report any original code. We reanalyzed datasets EGAS00001001732 and GenBank: GSE118014 (see [key resources table](#)).
- The mass spectrometry proteomics data have been deposited in the ProteomeXchange Consortium via the PRIDE⁵⁷ partner repository with the dataset identifier PRIDE: PXD065389.
- Additional information required to reanalyze the data reported herein is available from the [lead contact](#) upon reasonable request.

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Figure 7. The *SRRM3*-regulated microexon program controls the increase of the neural character of PanNETs

- (A) Heatmap of the expression levels of genes expressed specifically in neural cells. Rows represent the genes, and columns represent the samples analyzed.
- (B) Bar plots of RT-qPCR analyses of neural specific genes (*TUBB3* and *PGP9.5*) in RNAs extracted from the pancreata of the RT2 mice and the respective WT at the indicated time points.
- (C) Tumor sections from mouse No.1, derived from both NT and Sh1 cells, were stained with hematoxylin. Immunohistochemistry was performed using anti-TUBB3 antibodies, with representative images and corresponding negative controls shown.
- (D) Bar plots of RT-qPCR analyses of *TUBB3* and *PGP9.5* in RNAs extracted from mice from the xenograft experiment described in [Figure 7](#).
- (E) Plots of RT-qPCR analyses of neural specific genes as indicated, in QGP-1 cells treated with *SRRM3* siRNA or a pool of three ASOs. For B, D, and E, GAPDH levels were used for the normalization. Error bars represent $\pm$ SEM. ns, $p > 0.05$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$.

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AUTHOR CONTRIBUTIONS

M.P. performed the bioinformatics analyses, cellular assays, animal studies, and *in vitro* splicing analyses; generated figures; and contributed to writing the manuscript. C.M. performed the majority of the splicing analyses and contributed to the animal and cellular studies. Z.E. performed the RINS1 experiments and prepared Figures 6A, 6B, and S6A. G.R. and V.K. advised on image acquisition and analyses and prepared Figures 6C and S6B. A.K. performed RNA preparations from animals and cells and islet isolation from mouse pancreas. A.M. performed the xenograft animal studies. M.D. and V. N. performed the ultrasound imaging of the xenograft tumors and prepared Figures 7E and S7B. A.K. prepared the samples for the mouse insulin ELISA assays. J.S. provided the PanNET cell lines and advised on performing the cellular assays. J.J.-M. provided advice on islet experimentation and provided ASOs for splicing experiments. S.G.D. advised and contributed to the calcium experiments and the preparation of Figures 5 and S5. M.S. performed the LC-MS/MS experiment and data analysis. M.E.R. performed part of the bioinformatics analyses and cellular assays, organized the bioinformatics analyses, and supervised M.P. on the bioinformatics analyses. P.K. conceived, designed, and supervised the study; wrote the manuscript; and prepared and supervised the preparation of the figures. All authors reviewed the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Insulin Antibody	Abcam	Cat# ab181547;RRID:AB_2716761
TBB3 Antibody	Sigma Aldrich	Cat# T2200; RRID:AB_262133
HRP-conjugated Goat Anti-Rabbit IgG	Santa Cruz Biotechnology	Cat# sc-2004; RRID:AB_631746
CoraLite® Plus 647-conjugated GOLGA2/GM130 Antibody	Proteintech	Cat# CL647-11308; RRID:AB_2934854
Beta-tubulin Monoclonal Antibody	Abcam	Cat# ab52623; RRID:AB_869991
Anti-mouse Alexa Fluor 488	Molecular Probes	Cat# A28175; RRID:AB_2536161
Goat anti-GFP Antibody	SICGEN Antibodies	Cat# AB0020; RRID:AB_2333100
Goat anti-mCherry Antibody	SICGEN Antibodies	Cat# AB0040; RRID:AB_2333093
HRP-conjugated Donkey anti-Goat IgG	Santa Cruz Biotechnology	Cat# sc-2020;RRID:AB_631728
Bacterial and virus strains		
psPAX2 (Packaging Vector)	Addgene	Plasmid #12260
pMD2.G (Envelope Plasmid)	Addgene	Plasmid #12259
RINS1 Plasmid	Addgene	Plasmid #107290
Chemicals, peptides, and recombinant proteins		
RPMI 1640 medium	Gibco	Cat# 31870-025
RPMI 1640 GlutaMAX™ Supplement	Gibco/Thermo Fisher	Cat# 61870010
DMEM/F-12 (1:1) (1X)	Gibco	Cat# 11320-074
Opti-MEM® (1X)	Gibco	Cat# 11058-021
Thrypsin	Biosera	Cat# LM-T1705/100
Fetal bovine serum (FBS)	Corning	Cat# P40-57500
L-Glutamine+	Thermo Fisher Scientific	Cat# A2916801
Penicillin-streptomycin	Gibco	Cat# 15070-063
Epidermal growth factor (EGF)	Thermo Fisher Scientific	Cat# Peprotech100-15
Fibroblast growth factor 2 (FGF2)	Thermo Fisher Scientific	Cat# Peprotech100-18B
Collagen IV	Sigma Aldrich	Cat# C7521-5MG
TRIzol® Reagent	Thermo Fisher Scientific	Cat#15596026
Random hexamer primers	Invitrogen By Thermo Fisher	Cat# N8080127
Oligo(dT) primers	New England BioLabs	Cat# S1316S
RNaseOUT™ Recombinant Ribonuclease Inhibitor	Invitrogen By Thermo Fisher Scientific	Cat# 10777-019
Protoscript II Reverse Transcriptase	New England BioLabs	Cat# M0368L
KAPA SYBR® FAST qPCR mix	KAPA Biosystems	Cat# KK4602 07959397001
Matrigel	Corning	Cat#354230
Formalin	Alfa Aesar	Cat# 43368
BSA (Bovine Serum Albumin)	BIOFROXX	Cat# 1126GR100
Hematoxylin	Sigma Aldrich	Cat# 1.09253/4
SRRM3-siRNA (SMARTPool)	Dharmacon	Cat# L-016790-02-0005
Scramble-siRNA	Origene	Cat# SR30004
Lipofectamine RNAiMAX	Thermo Fisher Scientific	Cat# 1000014472
Lipofectamine 2000	Invitrogen	Cat# 52877
jetPRIME® Reagent	PolyPlus	Cat# 01000046 & 101000015
Puromycin	InvivoGen	Cat# ant-pr

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Hexadimethrine Bromide (Polybrene)	Sigma Aldrich	Cat# H9268
Fluo-4 a.m.	Thermo Fisher Scientific	Cat# F14201
HEPES-Buffered Saline (HBS)	Fisher Scientific	Cat# BP310-100
Pluronic-F127 (20% w/v in DMSO)	Thermo Fisher Scientific	Cat# P3000MP
Bovine Serum Albumin (BSA)	Sigma-Aldrich	Cat# A9418
Triton X-100	Fisher Scientific	Cat# 10591461
BAPTA	Sigma-Aldrich	Cat# 196418
Nifedipine	Sigma-Aldrich	Cat# 21829-25-4
MTT Reagent	Applchem	Cat# A2231,0001
DMSO	PAN Biotech	Cat# P60-36720100
KCl	Merck	Cat# 7447-40-7
cComplete™, Mini Protease Inhibitor Cocktail	Roche	Cat# 11836153001
Methanol (ice-cold)	CARLO ERBA Reagents	Cat# 524102

Critical commercial assays

RNeasy Micro Kit	Qiagen	Cat#74004
ELISA BrdU Kit	Roche	Cat# 11647229001
Mouse Ultrasensitive Insulin ELISA kit	ALPCO	Cat# 80-INSMSU-E01, E10
Human Insulin ELISA kit	ALPCO	Cat# 80-INSHU-E01.1
Corticosterone ELISA kit	CAYMAN CHEMICAL COMPANY	Cat# 501320
DAB Substrate Kit	VECTOR LABORATORIES	Cat# SK-4100
Transwell assay	Costar	Cat# 30014004
PCR Mycoplasma Detection Kit	ABM	Cat# ABM-G238

Deposited data

PanNET public RNA sequencing dataset	Scarpa et al. ¹⁸	EGAS00001001732
PanNET public RNA sequencing dataset	Chan et al. ¹⁷	GSE118014
Whole pancreatic islets RNA sequencing dataset	Tapial et al. ¹⁶	GSE50398
Human adenocarcinoma dataset	Kirby et al. ⁵⁸	GSE79668
Proteomics dataset	This study	PXD065389

Experimental models: Cell lines

QGP-1 cell line		RRID:CVCL_3143
NT-3 cell line		RRID:CVCL_VG81
293T/LentiX Cells	Takara Bio	Cat# 632180
BON-1		RRID:CVCL_3985

Experimental models: Organisms/strains

Mouse: C57Bl6/J	The Jackson Laboratory	JAX:000664
Mouse: NOD.CB17-Prkdc ^{scid} /J	The Jackson Laboratory	JAX:001303

Software and algorithms

ImageJ Software	NIH	v1.53t
VevoLAB Software	FUJIFILM VisualSonics	v5.8.1.3266
SoftMax Pro 7	Molecular Devices	5049092 E
GraphPad Prism (v8)	GraphPad	8.0.2
Magellan™ Software	TECAN life sciences	N/A
Imaris Software	Oxford instruments	N/A
DIA-NN, v1.8.1	Demichev et al., 2020 ¹⁷	N/A
Perseus, v2.0.5.0	Tyanova et al., 2016 ¹⁸	N/A
BioRender	https://www.biorender.com	N/A
Vast-tools	Tapial, J. et al., (2017) ⁴	v2.5.1
STAR	Dobin, A. Et al., (2013) ⁶	vSTAR-2.7.10a

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Matt software	Gohr, A., & Irimia, M. (2019) ¹⁰	v1.3.0
R	RCore Team (2022) ¹⁵	v4.2.1
R studio	RStudio Team (www.rstudio.org/)	N/A
VastDB	Tapial, J. et al., (2017) ⁴	v1.0
iDEP	Ge, S.X., Son, E.W. & Yao, R. (2018) ⁷	v2.0.1
PanglaoDB	Oscar Franzén, Li-Ming Gan, Johan L M Björkegren (2019) ⁹	N/A
ggplot2	Wickham et al. (2016) ¹⁹	v3.4.2
ggfortify	Tang et al. (2016) ²⁰	v0.4.16
pheatmap	Kolde R. (2018)	v1.0.12
ggpubr	Kassambara, A. (2022)	v0.6.0
corrplot	Wei et al. (2017)	v0.95
revigo	Supek et al. (2011)	v1.8.0
clusterProfiler	Yu et al. (2012)	v4.2.1
rstatix	Kassambara et al. (2020)	v0.7.2
matrixTests	Petersen et al. (2017)	v0.2
RColorBrewer	Neuwirth et al. (2014)	v1.1.3

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

The mouse model RIP1-Tag2 in the C57Bl6/J background was used for validation of the selected alternative splicing events. Three timepoints (T1, T2, T3) of mice growth were used. Sex selection was randomized. RIP1-Tag2 mice start to develop beta-cell hyperplasia at around 8 weeks of age (T1) and they normally die between 12 and 14 weeks of age (T2). Upon high-carbohydrate diet, they can survive until 17–18 weeks along with developing highly invasive carcinomas (T3).⁵⁹ The mice were euthanized and either islets were extracted for RNA or tumors/total pancreases were dissected for RNA isolation and paraffin embedding for subsequent analyses. In total, 35 wild type (WT) (C57Bl6/J) and 35 transgenic (RT2 in C57Bl6/J background) mice were used for validations. Animals were fed with a standard laboratory chow and 5% sugar water abundance *ad libitum*.

For the mouse xenografts study, 1×10^6 cells suspended in 100 μ L of PBS:Matrigel (1:1; Corning) of QGP-1 NT and QGP-1 Sh1 or Sh2 cells, were injected into the left and right flank of 15-week-old NOD-SKID mice (NOD.CB17-Prkdcscid/J, Charles River, Strain code: 634), respectively. A total of 9 mice were used and 3D ultrasound images (via 3D scans) were acquired using the Vevo 3100 Imaging System (FUJIFILM VisualSonics). Tumor growth was monitored for up to 4 weeks and weekly calculations were carried out using the VevoLAB software (version 5.8.1.3266). At the endpoint, mice were euthanized, and both tumors and pancreata from each mouse were weighed and collected for RNA isolation and paraffin embedding for subsequent analyses. All animals were maintained in a climate-controlled environment at 21°C–22°C and 50%–60% humidity with a light dark cycle (12:12h) and standard laboratory chow.

Ethical approval

All studies utilizing mice were approved by the Institutional Committee of Protocol Evaluation in conjunction with the Veterinary Service Management of the Hellenic Republic Prefecture of Attika according to all current European and national legislation and were performed in accordance with relevant guidelines and regulations under the relevant animal protocol licenses with numbers 278206-01/04/2022, 448199-06/08/2019. Experiments were performed in accordance with the guidance of the Institutional Animal Care and Use Committee of BSRC ‘Alexander Fleming’.

Cell lines

The human pancreatic cancer cell lines QGP-1, BON-1 and NT-3 were used for this study. QGP-1 (RRID:CVCL_3143) is a neuroendocrine tumor cell line of pancreatic somatostatinoma origin. QGP-1 were cultured in RPMI1640 medium (GIBCO Cat# 31870-025) supplemented with 10% fetal bovine serum (FBS), 1% L-Glutamine+ and 1% penicillin–streptomycin. The NT-3 (RRID:CVCL_VG81) cell line originated from a pancreatic neuroendocrine tumor G2 stage, were cultured under semi-adherent conditions in collagen IV-coated plates (Sigma C7521-5MG, final concentration 50 μ g/ml) in RPMI 1640 GlutaMAX Supplement (Gibco/ThermoFisher, Cat#61870010) accompanied with 10% FBS, penicillin/streptomycin, epidermal growth factor (EGF, 20 ng/mL), and fibroblast growth factor 2 (FGF2, 10 ng/mL). The BON-1 cell line is a well-established human pancreatic neuroendocrine tumor cell line derived

from a lymph node metastasis of a pancreatic carcinoid tumor. BON-1 were cultured in DMEM/F-12 (1:1) (Gibco, Cat# 11320-074) supplemented with 10% fetal bovine serum (FBS), 1% L-Glutamine+ and 1% penicillin–streptomycin. The cells were maintained in a standard humidified incubator at 37°C in a 5% CO₂ atmosphere. All three cell lines were not authenticated; however, they were regularly tested negative for mycoplasma contamination using the PCR Mycoplasma Detection kit from ABM.

METHOD DETAILS

Alternative splicing and differential gene expression analyses

Access was gained to publicly available RNA sequencing datasets from normal pancreas and from PanNET patients. In detail, two datasets were analyzed as normal pancreatic tissue and two as PanNET. PanNET public RNA sequencing datasets can be found under accession numbers EGAS00001001732 and GSE118014.^{17,18} Normal pancreatic tissue datasets derived from VastDB repository for each different type of pancreatic cell and under accession number GSE50398,¹⁶ for whole pancreatic islets. At the same time, a publicly available human adenocarcinoma dataset was also analyzed for splicing to use for comparisons with PanNET samples, with accession number GSE79668.⁵⁸ FASTQ reads were aligned with the human VASTDB library (hg38, vastdb.hs2.23.06.20).

The code used for splicing analysis was based on the SpliceNet repository (<https://github.com/estepi/SpliceNet>) and was modified for this study as needed. Alternative splicing analysis was performed with VAST-TOOLS v2.5.1¹⁶ and results were depicted as changes in percent-spliced-in values (Δ PSI). PSI values for single sample replicates were quantified for all types of alternative events, comprising single and complex exon skipping events (S, C1, C2, C3, ANN), microexons (MIC), alternative 5' and 3' splice sites (Alt5 and Alt3, respectively) and retained introns (IR-S, IR-C). Events showing splicing change $|\Delta$ PSI| > 15 and P-Adjusted value < 0.05 calculated using non-parametric Kruskal-Wallis test (R library matrixTests, v0.2), followed by Dunn's Test (R library rstatix, v0.7.2), were considered preferred regulated events under tumoral conditions.

For differential gene expression analysis, FASTQ reads were aligned to the GRCh38 human reference genome assembly using the STAR alignment tool (vSTAR-2.7.10a).⁶⁰ The aligned reads were then used to generate BAM files and extract the gene reads for downstream analyses. iDEP workflow was used to generate Log₂FoldChange values between control and PanNET samples.⁶¹ Genes showing $|\text{Log}_2\text{FoldChange}| > 0.58$ and P-Adjusted value < 0.05 were considered differentially expressed.

ORF impact prediction

Potential ORF impact of alternative exons was predicted as described by Irimia et al.²⁴ Exons were mapped on the coding sequence (CDS) or 5'/3' untranslated regions (UTR) of genes. Events mapping on the CDS were divided into CDS-preserving or CDS-disrupting.

Cell markers

Utilizing the PanglaoDB database,⁴⁴ specific cell markers were selected and downloaded regarding normal pancreatic tissue, including alpha, beta, pancreatic progenitor, stellate and peri-islet Schwann cells. The retrieved cell marker data was used to generate a heatmap plot illustrating the expression patterns across our samples according to the color scale employed. Visualization was made using pheatmap (v1.0.12) package in R, with rows representing cell types and columns representing our samples.

RNA maps analysis

To generate the RNA maps, we used the rna_maps function (Matt software v1.3.0⁶²), using sliding windows of 11 nucleotides. Searches were restricted to the affected exons and the first 75 nucleotides of the upstream intron. Regular expression was used to search for the binding motif of SRRM3 (UGC). RNA maps for the SRRM3 motif were analyzed. Cassette exons were grouped as follows: up Δ PSI > 15 and PSI margin between groups > 5, down Δ PSI < -15 and PSI margin between groups > 5. The sequence of first 15 nts of exons and the first 75nt of introns (sliding window = 11, p value ≤ 0.05 with 1000 permutations) were compared with the non-changing exons ($\text{ndiff} -2 > \Delta$ PSI > 2 and average PSI controls < 95 and Δ PSI ≤ 5).

Gene ontology analysis

Ensembl gene IDs for the significant MIC splicing events (Length < 50) were analyzed using clusterProfiler (v4.2.1) R package. Statistical significance was defined with p -Adjusted value < 0.05 in the comparison of PanNET to islets and the positive false discovery rate with q value < 0.05. The obtained GO annotations for our genes were further reduced using revigo R library (rvgo, v1.8.0), to identify enriched functional categories.

Correlation analyses

For comparison with cancer samples from the Cancer Genome Atlas Program (TCGA), lists of significantly deregulated splicing events between tumors and normal tissues ($|\Delta$ PSI| > 5, p value < 0.05) were obtained from the analysis done by Head et al. (2021).⁶³ Using Pearson coefficient method and the corplot R library (v0.92), the association between altered exonic splicing events in PanNET in comparison to islets ($|\Delta$ PSI| > 15, P-Adjusted value < 0.05) and deregulated events in other types of cancer, was evaluated. For the correlation thresholds, $|\text{association value}| > 0.6$ and FDR < 0.05 were applied.

For the correlation between splicing factors and splicing events, again the corplot R library with Pearson coefficient was used. The splicing factors with $|\text{Log}_2\text{FoldChange}| > 0.58$ and P-Adjusted value < 0.05 emerging from the differential expression analysis, were

associated with the deregulated exonic splicing events in PanNET compared to islets ($|\Delta\text{PSI}| > 15$, P-Adjusted value < 0.05), applying $\text{fdr} < 0.05$.

Identification of common splicing events between datasets

In order to support our findings regarding elevated SRRM3-regulated splicing events in PanNET, common splicing events between our splicing analysis and a list of downregulated splicing events upon SRRM3-KD in EndoC-BH1 cells were identified and visualized. In addition, the same method was applied with a list of downregulated splicing events upon SRRM3-KD in rat INS-1E cells. Both lists were downloaded from Juan-Mateu et al.²³

SRRM3 gene expression plots

SRRM3 $\log_2\text{FoldChange}$ values were acquired from Head et al. (2021)⁶³ and visualized in R, using ggpubr library (v0.6.0) in comparison to PanNET. In addition, individual cRPKM values for SRRM3 gene were downloaded from VastDB repository for distinct types of normal tissues, including cerebellum, retina, whole brain, cortex, adrenal, heart, endometrium, prostate, adipose, bladder, thyroid, breast, melanocytes, lung, kidney, endothelium, muscle and liver, to compare them with SRRM3 expression in PanNET and the normal pancreatic tissues we analyzed.

Glucose measurements

Glucose levels were measured using the Contour Care glucometer in 8, 12 and 16 weeks mice. For glucose measurements collected blood from 22 mice in total. For 8 weeks timepoint blood was collected from 3 WT to 5 TG mice, while for 12 and 16 weeks timepoints blood was collected from 7 WT to 7 TG mice.

Corticosterone ELISA protocol

For measuring corticosterone levels in mouse samples, blood from 8, 12 and 16 weeks mice was collected and the serum of each sample was isolated. Each sample was diluted 1:250 and processed according to the instructions of the manufacturer (CAYMAN CHEMICAL COMPANY, cat# 501320). In each timepoint 3 WT and 3 TG samples were used, inserted as triplicates in the kit plate.

Insulin ELISA protocol

For measuring insulin levels in human and mouse samples, supernatant from NT3 cells and blood from 8, 12 and 16 weeks mice were collected. For mouse ELISA kit in each timepoint 3 WT and 3 TG samples were used, inserted as triplicates in the kit plate. For the blood samples, serum was then isolated through centrifuge. The samples were used undiluted according to the instructions of the manufacturer (ALPCO, cat# 80-INSHU-E01.1, cat# 80-INSMSU-E01, E10).

RNA isolation, cDNA synthesis, and PCR/qPCR

Total RNA from murine pancreas was extracted using RNeasy Mini Kit (Qiagen) following the manufacturer's instructions. RNA isolation from cell lines was performed with the TRIzol reagent (Thermo Fisher Scientific, Inc). Islet isolation and subsequent RNA extraction was carried out as described by Li et al..⁶⁴

Reverse transcription (RT) was carried out with 400 ng to 1 μg of total RNA, using random hexamer primers (Invitrogen), OligodT (New England BioLabs, Inc.), RNaseOUT Recombinant Ribonuclease Inhibitor (Thermo Fisher Scientific) and Protoscript II reverse transcriptase (New England BioLabs, Inc.), according to manufacturer's instructions.

Semi-quantitative PCR reactions were carried out with 2 μL of cDNA, with different cycling and temperature parameters for each selected alternative splicing event (Table S5). Inclusion/exclusion bands for these events were analyzed by acrylamide electrophoresis and quantified using ImageJ (v1.53t). Quantitative PCR (qPCR) was performed in a Corbett Rotor-Gene RG 6000 using Kapa SYBRFAST (KAPA Biosystems) qPCR mix.

Immunohistochemistry and H&E staining

After pancreases were obtained from RIP1-Tag2 mice or at the endpoint of the xenograft experiment, parts of the pancreases or tumors were dissected, fixed in formalin and embedded in paraffin. Sections were cut at 5 μm thickness, deparaffinized, re-hydrated in descending ethanol grades, washed and boiled at 95°C for 15 min in sodium citrate buffer pH 6.0 for antigen retrieval. Blocking was performed with 5% BSA for 1 h and sections were then incubated with the following primary antibodies overnight at 4°C diluted in BSA: Insulin (Cat #EPR17359, 1:200), TBB3 (Cat #T2200, 1:200). Sections were subsequently washed and incubated in 3% H_2O_2 solution for 10 min. HRP-conjugated goat anti-rabbit IgG (Cat # sc-2004, 1:5000) was applied for 1h at RT and the DAB Substrate Kit was used to visualize the signal. The sections were counterstained with hematoxylin and images were obtained with a NIKON Eclipse E600 microscope, equipped with a Qcapture camera.

Transfection of BON-1, QGP-1, and NT-3 cells

For RNA-mediated interference, cells were transfected with control (scramble) or SRRM3-siRNA (L-016790-02-0005 SMARTPool siRNA) at 30 nM final concentration for 72 h, using the Lipofectamine RNAiMAX transfection reagent (Thermo Fisher Scientific), according to manufacturer's instructions. Alternatively, NT-3 cells were transfected with control or SRRM3-siRNA by electroporation using the

Super Electroporator NEPA21 (Nepa Gene Co., Ltd; Poring pulse: pulse voltage 175 V; pulse interval 50 ms; pulse width 5 ms; pulse number 2; and Transfer pulse: pulse voltage 20V; pulse interval 50 ms; pulse width 50 ms; pulse number 5). For co-transfection of NT-3 cells with siRNAs and RINS1 plasmid (gift from Dmytro Yushchenko, Addgene plasmid # 107290, <https://doi.org/10.1016/j.chembiol.2017.03.001>), lipofectamine 2000 transfection was used (Thermo Fisher Scientific), according to manufacturer's instructions.

For antisense oligonucleotides (ASOs) transfection, QGP-1 and BON-1 cells were transfected with control (scramble) siRNA or ASO targeting three individual microexons (CADPS2, DADNA1D, PTK2) at 40nM final concentration for 72h using the Lipofectamine RNAiMAX transfection reagent (Thermo Fisher Scientific), according to manufacturer's instructions. ASOs were designed as described in by Juan-Mateu et al.,²¹ based on the following sequences: UUACUCACCUCUCUCCCCUGA for CADPS2, GGGUUCUAUCCAGGAAAGGCA for CACNA1D and CCAUAGCUUUCUGUAUUGCAA for PTK2 targeting.

For viral transduction in QGP-1 and BON-1 cells, the sequences of shRNA used are: Sh1 (sh targeting SRMM3 CGTTACGAACACAG GAATCTC) and Sh2 (sh targeting SRMM3 GCATAGGATAACATGTGCTTT). Lentiviral vector production and transduction were performed using standard protocols. Individual clones were selected with puromycin (final concentration 0.5μg/ml) and kept in cell culture.

Lentivirus production

4×10^6 293T/LentiX cells were seeded per 10cm dish under semi-adherent conditions in collagen IV-coated plates (Sigma C7521-5MG, final concentration 50μg/ml). After 24h at the incubator, a mix of transgene plasmid (NT or Sh), viral packaging vector (psPAX2) and viral envelope plasmid (pMD2G) was added in 500μL Jetprime buffer at a 2.8:1.8:1 DNA ratio. The mix was left at RT for 5min, before 16μL jetPRIME reagent was added. The transfection mix was then transferred in a falcon and incubated for 20min at RT. 6.5mL DMEM medium (Lonza, Cat#12-733F), supplemented with 10% fetal bovine serum (FBS), 1% L-Glutamine+ and 1% penicillin-streptomycin, were added to the falcon and replaced the culture medium. After 6h, the transfection mix was replaced with fresh DMEM medium. Three days after transfection, culture medium was selected and the virus suspension was filtrated through a sterile 0.45μm filter, aliquoted and stored at -80°C for further use.

Lentiviral transduction

2.5×10^5 QGP-1 or BON-1 cells per well were seeded in a 6-well plate with RPMI1640 medium (GIBCO Cat# 31870-025) supplemented with 10% fetal bovine serum (FBS), 1% L-Glutamine+ and 1% penicillin-streptomycin. After 24h, the filtered lentiviral supernatant along with polybrene (8μg/ml final concentration) were added to the culture medium at a 1:1 ratio. The next day the transduction mix was replaced with fresh RPMI1640 and left for 24h in the incubator. Positive clones were then selected using puromycin antibiotic (1μg/ml final concentration). Once stable, puromycin-resistant cells were tested for SRRM3 downregulation using qPCR and individual clones were generated.

Cell proliferation assays

In 96-well plates, 7.5×10^3 QGP-1 or BON-1 cells/well were placed and allowed to grow for 7 days at 37°C with 5% CO_2 . Cell proliferation was assessed using the ELISA BrdU Kit (Roche, Cat. No. 11647229001) according to the manufacturer's instructions. Absorbance was measured at 370 nm with a reference wavelength of 492 nm using the OptiMax microplate reader (Molecular Devices). For cell proliferation measurements during ASO transfection, 20×10^3 cells were seeded.

Intracellular calcium measurements

QGP-1 and NT-3 cells were seeded in 96-well cell culture plates at a confluency of 30000 cells/well and the following day were assayed for $[\text{Ca}^{2+}]_i$ measurements using a FlexStation 3 96-well multi-mode microplate reader (Molecular Devices). Before the assays, cells were loaded with Fluo-4/AM (F14201, ThermoFisher Scientific) in 100 μL of HEPES-Buffered Saline (HBS:135 mM NaCl, 5.9 mM KCl, 11.6 mM HEPES, 1.5 mM CaCl_2 , 1.2 mM MgCl_2 , 1 mM glucose (pH 7.3) that contained 0.02% Pluronic-F127 (P3000MP, ThermoFisher Scientific, 20% (w/v) solution in DMSO) and supplemented with 0.1% Bovine serum albumin (A9418, Sigma-Aldrich). Cells were incubated for 1 h at 37°C , then the medium was removed and replaced with HBS, supplemented with 0.1% Bovine serum albumin, for 45 min at 37°C before the assays. Measurements were taken at 25°C on the FlexStation 3 with reads taken at 1.5 s intervals at Exc./Em. 490/520 nm wavelengths, in FLEX mode using medium photomultiplier tube (PMT) gain and 5 flashes per read. Cells were assayed in a volume of 180 μL before the addition of 20 μL 0.5 M KCl or 20 μL 0.2 M Glucose in each well at 120 s after the onset of measurements. Assays were carried out for 12 min and then measurements of F_{max} and F_{min} were taken by the addition of Triton X-100 (10591461, Fisherscientific)(2% final concentration) and BAPTA (196418, Sigma-Aldrich) (10 mM final concentration), sequentially and at 1 min intervals on each well upon completion of measurements. Results were visualized with the SoFtMaxPro© 7 software (Molecular Devices) and fluorescent intensities were converted to $[\text{Ca}^{2+}]_i$ concentrations using the equation $[\text{Ca}^{2+}]_i \text{ (nM)} = K_d \times (F - F_{\text{min}} / F_{\text{max}} - F)$ where K_d for Fluo-4 is 335 nM. When Nifedipine was used, it was added in each well offline 3 min before the start of fluorescence measurements to give a final concentration of 10 μM in each well. At least 4 biological replicates were carried out with 6 wells assayed for each condition in each biological replicate. Data on $[\text{Ca}^{2+}]_i$ was visualised in GraphPad Prism (v8) with error bars representing $\pm$ SEM.

MTT cell viability assay

24h after transfection, QGP-1 cells were plated at a density of 5000 or 20000 cells/well in triplicates on a 96-well plate, for nifedipine treatment or siRNA/ASO transfection. At 48h or 72h post-transfection, cell viability was assessed using MTT reagent (Applichem) dissolved in PBS (5mg/ml). On the day of measurement, fresh medium containing diluted MTT (1:10, 0.5mg/ml final concentration) was added and incubated for 3 h at 37°C. After removing the medium, formazan crystals were dissolved in 200μL solution of DMSO. MTT reduction was quantified by measuring the absorbance at 570 nm using the Tecan Sunrise microplate reader and the Magellan software.

Analysis of insulin secretion with the ratiometric reporter RINS1

The protocol described in Schifferer et al.³⁸ was used, with small modifications. Briefly, NT-3 cells were co-transfected with control or *SRRM3*-siRNA and RINS1 plasmid. After 72 h, the medium was replaced with low glucose containing Imaging buffer (115 mM NaCl, 1.2 mM CaCl₂, 1.2 mM MgCl₂, 1.2 mM K₂HPO₄, 11.1mM glucose, 20 mM HEPES pH 7.4) and the cells were allowed to adapt for 2 h at 37°C. Insulin secretion was stimulated by exposing cells to Imaging buffer supplemented with 20 mM glucose and 30 mM KCl. Supernatants were collected after 0- and 10-min and cleared by centrifugation at 600 x g prior to measurement of sfGFP and mCherry fluorescence intensities. Cells were lysed with lysis buffer (10 mM Tris-HCl pH 7.4, 150 mM NaCl, 0.5 mM EDTA, 0.5% NP-40) supplemented with protease inhibitors, and diluted 3-fold in Imaging buffer prior to fluorescence measurement. sfGFP and mCherry fluorescence intensities were measured on a FlexStation 3 Multi-Mode Microplate reader system and SoftMax Pro 6 software for analysis with the following settings: for sfGFP, Excitation 485 nm; Cut off 515 nm; Emission 520 nm; for mCherry, Excitation 587 nm; Cut off 610 nm; Emission 620 nm. Insulin secretion was estimated by the ratio of the sfGFP versus the mCherry signal intensity in the supernatant, after normalization against the respective signals in the lysate.

Immunostaining with golgi markers

Cells were fixed for 5 min at -20°C with ice-cold methanol. Rabbit CoraLite Plus 647-conjugated GOLGA2/GM130 polyclonal antibody (Proteintech, Cat # CL647-11308, 1:100) and mouse beta-tubulin monoclonal antibody (1:100) were used as primary antibodies. Anti-mouse Alexa Fluor 488 (Molecular Probes Cat# A28175) was used as secondary antibody. Analysis of *cis*- Golgi area relatively to cell surface was performed with ImageJ (v1.53t). Analysis of Golgi area relatively to nuclei was performed with Imaris software.

Western blot analysis

Goat polyclonal anti-GFP and anti-mCherry (SIGGEN Antibodies, Cat # AB0020 and AB0040, respectively, 1:1000) were used as primary antibodies. HRP-conjugated donkey anti-goat IgG (Santa Cruz Biotechnology, Cat # sc-2020) was used as secondary antibody.

Migration assay

In order to examine cell motility, migration assay with transwells was used. Inserts were pre-coated with collagen IV for 1h. BON-1 (transfected with either NT or Sh1) and QGP-1 (NT/Sh1) cells were starved with Opti-MEM for 2h prior to the plating. In 96-well plate, 5×10⁴ cells/well in Opti-MEM were placed inside the insert and complete medium was added to the bottom of each well. Then cells were allowed to migrate for 16h at 37°C with 5% CO₂.

After the completion of the 16h, Opti-MEM was aspirated from the insert and non-migrated cells were removed using a cotton swab. The membrane was then washed gently with PBS. The migrated cells on the bottom of the insert were fixed with 4% paraformaldehyde (PFA) for 10 min and then stained with DAPI (1:500) for 1h for visualization. Each stained insert was examined under Axio Observer A1 (ZEISS) microscope with automated stage and the migrated cells were counted in multiple fields (8 random fields). The final measures were calculated using ImageJ (v1.53t), incorporating the area of the insert and the microscope scale.

Invasion assay

In order to examine matrix invasion, invasion assay with transwells was used. Inserts were pre-coated with Matrigel (1:1 dilution with medium) for 1h. BON-1 (transfected with either NT or Sh1) and QGP-1 (NT/Sh1) cells were starved with Opti-MEM for 2h. In 96-well plate, 5×10⁴ cells/well in Opti-MEM were placed inside the insert and complete medium was added to the bottom of each well. The cells were allowed to invade for 16h at 37°C with 5% CO₂. Triplicates were used for each condition and cellular type.

After the completion of the 16h, Opti-MEM was carefully aspirated from the insert and non-invaded cells were removed using a cotton swab. The membrane was then washed gently with PBS. The invaded cells on the bottom of the insert were fixed with 4% paraformaldehyde (PFA) for 10 min and then stained with DAPI (1:500) for 1h for visualization. Each stained insert was examined under Axio Observer A1 (ZEISS) microscope with automated stage and the invaded cells were counted in multiple fields (8 random fields). The final measures calculated using ImageJ (v1.53t) incorporated the area of the insert and the microscope scale. Triplicates were used for each condition and cellular type.

Mass spectrometry - Sample preparation

QGP-1 and BON-1 cells transfected with control (NT) or *SRRM3* shRNA were placed in TC 100mm cell culture dishes at 37°C with 5% CO₂. Triplicates of each cell type and condition were used. The cell extract was collected and processed according to the Sp3 protocol including an alkylation step in the dark for 15 min in 10 mg/mL iodoacetamide. 20 μg of beads (1:1 mixture of hydrophilic and hydrophobic SeraMag carboxylate-modified beads, Cytiva) were added to each sample in 50% ethanol. The proteins were allowed

to bind to the beads for 5 min followed by repeated steps of protein clean-up using a magnetic rack. The beads were washed two times with 80% ethanol and once with 100% acetonitrile (Fisher Chemical). The captured beads proteins were digested overnight at 37°C under vigorous shaking (1200 rpm, Eppendorf Thermomixer) with 0.5 µg Trypsin/LysC (MS grade, Promega) prepared in 25 mM ammonium bicarbonate. The next day, the supernatants were collected and the peptides were purified using a modified Sp3 clean-up protocol and finally solubilized in the mobile phase A (0.1% formic acid in water), sonicated and the peptide concentration was determined through absorbance at 280 nm measurement using a nanodrop instrument.

Mass spectrometry - nanoLC- diaPASEF

The peptidic samples were analyzed in three technical replicates using a NanoElute 2 liquid chromatography system coupled to a timsTOF Ultra 2 mass spectrometer (Bruker Daltonik GmbH). Peptides were separated on an thermostated (50°C) evosep column (15 cm × 75 µm, 1.9 µm particle size, Evosep) using the captive spray Ultra 2 source and a captive spray stainless steel emitter and eluted over a 58-min gradient using a flow rate of loading and final equilibration with 350 nL/min while lowered to 200 nL/min in the separation phase (7–51 min). The mass spectrometer was operated in the dia-PASEF mode using the default method for low sample amount - 0.64–1.45 mobility and cycle time 0.96 s and ICC2 activated. Full MS data were acquired in the range of 100–1700 m/z. The ion mobility settings consisted of 3 isolation windows that were covered within a single TIMS scan where 8 TIMS scans were required to cover 24 windows.

Mass spectrometry - Data analysis

The timsTOF raw data was analyzed in DIA-NN 2.1.0 (Data-Independent Acquisition by Neural Networks) through searching against the reference Homo sapiens database and the DIA-NN contaminant database (total of 20 681 proteins, downloaded 20/3/2025) in the library free mode of the software, allowing up to two tryptic missed cleavages. The quantification strategy used was the newly introduced QuantUMS (Quantification using an Uncertainty Minimizing Solution) in high precision mode and the scoring was set to generic. A spectral library was created from the DIA runs and used to reanalyse them. DIA-NN default settings have been used with oxidation of methionine residues and acetylation of the protein N-termini set as variable modifications and carbamidomethylation of cysteine residues as fixed modification. N-terminal methionine excision was also enabled. The match between runs (MBR) feature was used for all analyses and the output (precursor and protein group) was filtered at 1% FDR.

The generated results were processed statistically and visualized in the Perseus software (1.6.15.0). The QuantUMS/MaxLFC protein group quantities (total of 6992) were uploaded and Log2 transformed and the proteins present with less than 70% valid values in at least one group. The remaining missing values were imputed using the default values in Perseus. T test results were visualized as Volcano plot in Perseus. Perseus-derived values were plotted using GraphPad Prism (v8).

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical tests were performed as indicated in figures legends using R (v4.2.1)⁶⁵ or GraphPad Prism (version 8). Principal Component Analysis (PCA) was performed on PSI values of the top 5000 most variable events and visualized using ggfortify (v0.4.16) and ggplot2 (v3.4.2) packages in R. Venn diagram was plotted using VennDiagram package in R (v1.7.3). For heatmaps PSI were plotted using pheatmap (v1.0.12) package in R. Hierarchical clustering was performed using a Ward's method and Euclidean distance as the distance metric.

Unless otherwise stated in the Figure legend, for statistically significant changes, two-tailed unpaired t test was used. ns, $p > 0.05$; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$. For the MS analysis, the t test result for SRRM3 were: for BON1, -Log(P-value): 2.19 and Difference (Log2fc): -1.28. For QGP1, -Log(P-value): 1.54 and Difference (Log2fc): -1.21. The protein Srrm3 was confidently identified with two proteotypic tryptic peptides: EGVLTREDRPGGHIVAETPR and VGTFRQMLMEK.